



GE Medical Systems

gemedical.com

Technical Publication

Direction 2382593-100

Revision 7

**GE Medical Systems
LightSpeed 5.X Pre-Installation:
LightSpeed¹⁶, Ultra & Plus**

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IMPORTANT PRECAUTIONS

LANGUAGE

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTES AVISOS PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

このサービスマニュアルには英語版しかありません。

GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。

このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。

この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意：

本维修手册仅存有英文本。

非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。

未详细阅读和完全了解本手册之前，不得进行维修。

忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent write "Damage In Shipment" on ALL copies of the freight or express bill BEFORE delivery is accepted or "signed for" by a GE representative or hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

Call Traffic and Transportation, Milwaukee, WI (262) 785 5052 or 8*323 5052 immediately after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage.

Complete instructions regarding claim procedure are found in Section S of the Policy And Procedures Bulletins.

14 July 1993

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

LITHIUM BATTERY CAUTIONARY STATEMENTS



CAUTION
Risk of
Explosion

Danger of explosion if battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer's instructions.



ATTENTION
Danger
d'Explosion

Il y a danger d'explosion s'il y a remplacement incorrect de la batterie. Remplacer uniquement avec une batterie du même type ou d'un type recommandé par le constructeur. Mettre au rebut les batteries usagées conformément aux instructions du fabricant.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the GEMS CQA Process to report all omissions, errors, and defects in this publication.

Revision History

Revision	Date	Reason for change
7	04/14/07	Chapter 2 - Added Information of Limited Access
6	08/12/05	Chapter 9 - Added CCC Compliance Note, to Figure 9-3.
5	10/01/04	Chapter 1 - Added EMC Information
4	09/10/04	Chapter 3 - Updated Option Catalog Number Chapter 6 - Updated Floor Loading Chapter 8 - Added 75kVA information.
3	01/28/04	Chapter 3 - Updated the table "System Options Catalog (Part) Numbers". Chapter 5 - Updated cooling requirements Chapter 9 - Added A1 panel cat num.
2	11/14/03	Chapter 9: Added the catalog number of the long cable set.
1	10/09/03	- Chapter 5 – Updated Table 5-1: Cooling Requirements (Worksheet) - General Updates
0	09/05/03	Initial release.

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Preface

Publication Conventions

Section 1.0

Safety & Hazard Information

1.1 Text and Character Representation

This publication uses various paragraph and character styles to indicate potential hazards. Key words "danger," "warning," "caution" and "notice" are used to identify important safety information. Text formatting (e.g., bold, all-caps, etc.) is applied to the paragraph contents to further highlight safety statements. Safety information normally identified in these hazard statements includes:

- Type of potential Hazard
- Nature of potential injury
- Causative condition
- How to avoid or correct the causative condition

A few examples are provided that have been adapted from GEMS' global document standard (2119696-100).



DANGER



USED WHEN A HAZARD EXISTS THAT WILL CAUSE SEVERE PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- ELECTROCUTION
- CRUSHING
- RADIATION



WARNING



USED WHEN A HAZARD EXISTS THAT COULD OR CAN CAUSE SERIOUS PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- POTENTIAL FOR SHOCK
- POTENTIAL FOR UN-COMMANDED MOTION



CAUTION



Used when a hazard exists that can or could cause minor injury to self or others if instructions are ignored. They include for example:

- Loss of critical patient data
- Sharp objects



NOTICE Equipment Damage Possible

Used when a hazard is present that can cause property damage but has absolutely no personal injury risk. They can include:

- Disk drive will crash
- Internal mechanical damage, such as to the x-ray tube
- Coasting the rotor through resonance.

It is important that the reader not ignore hazard statements in this document.

1.2 Graphical Representation

Important information is always preceded by an exclamation point contained within a triangle, . Additionally, several different icons (symbols) may be used to point out specific types of hazards that could possibly cause harm.

ELECTRICAL



LASER



MECHANICAL



HEAT



RADIATION



PINCH



Some others make you aware of specific procedures that should be followed.

AVOID STATIC ELECTRICITY



TAG AND LOCK OUT



WEAR EYE PROTECTION



Section 2.0 Conventions Used For General Information

2.1 Document Paragraph and Character Styles

Prefixes are used to highlight important non-safety related information. Paragraph prefixes (such as Example, Comment and Note) are used to identify important but non-safety related information. Text styles are also applied to text within each paragraph modified by the specific prefix.

Examples of prefixes used for general information:

Example: Used to make the reader aware that the paragraph(s) that follow are examples of information possibly stated previously.

Comment: Represents “additional” information that may or may not be relevant.

Note: Conveys information that should be considered important to the reader.

2.2 Page Layout

Headers and footers in this publication are designed to allow you to quickly identify your location. The document’s part number and revision number appear in every header on every page. Odd numbered page footers indicate the current chapter and page number. Even page footers show the current section and page number.

Chapter 1

Introduction

This direction contains physical and electrical data necessary for planning and preparing a site. Pre-installation work is defined as site preparation for installation of the GE CT scanner. It is the responsibility of the purchaser to arrange and pay for this work. Pre-installation work includes:

- Installation of electrical conduit, junction boxes, ducts, outlets, and line safety switches.
- Installation of interconnection wiring that is AWG stranded copper. The electrical contractor shall ring out and tag all wires at both ends. Color-coded wires are recommended for easier identification. Wires shall be continuous without splices. Ground wires must conform to local codes.
- Any site renovation.
- Alterations and modifications to products not specifically included in the sales contract.

All work must conform to local building and safety codes. Unless specifically mentioned, GE Medical Systems does not provide or install wires, conduits, junction boxes, and ducts as illustrated in this publication.

All CT site plans, preliminary concepts and final working drawings must be reviewed by General Electric Headquarters Architectural Planning prior to construction or approval.

Contact your local General Electric sales representative for complete information regarding your site-specific room layout.

Section 1.0

Site Readiness

The following **MUST** be completed before installation work can begin for a GE CT scanner:

- Finished wall, ceiling, floors & millwork
- Active phone line and network connection
- Power available to A1, w/provision for LOTO



NOTICE

An improperly prepared site (i.e., one that is in a state of construction) can result in *increased installation time*.

A CT scanner installed in a dirty environment is more prone to contamination, which can result in *decreased reliability* and *increased scanner downtime*.

Section 2.0

Responsibility of Purchaser

“Pre-installation” is work necessary to plan and prepare a site for installation of equipment. Avoid delay, confusion, and waste of manpower. Complete the following pre-installation work:

- Procure required materials.
- Install required material before delivery of system components.
- Complete all alterations and modifications if not specifically included in the sales contract.
- Complete the Pre-installation Check List in [Chapter 2](#).

Section 3.0

Site Preparation Prior to Equipment Delivery

- Determine room dimensions and verify entry doors are large enough to bring scanner system through.
- Install table/gantry floor duct.
- Install junction boxes of correct size with covers at locations shown in installation plan.
- Install conduit, duct, and raceway.
- Install power supply of correct voltage output and adequate KVA rating. Install local disconnects, including proper over-current protection.
- Install “Unistrut” or other suitable support work for mounting equipment on walls or from ceiling.
- Camera should be on-site at the time of installation.

Section 4.0

Medical Electrical Equipment for EMC

4.1 General Scope

This equipment complies with IEC60601-1-2 Edition 2 EMC standard for medical electrical equipment.

The **EQUIPMENT and/or SYSTEM** System is suitable to be used in the electromagnetic environment, as per the limits & recommendations described in the tables hereafter:

- Emission Compliance level & limits (see [Table 4-1](#))
- Immunity Compliance level & recommendations to maintain equipment clinical utility (see [Table 4-2](#), [Table 4-3](#), and [Table 4-4](#)).

Note: This system complies with above mentioned EMC standard when used with supplied cables up to maximum lengths referenced in the MIS MAPS or system cable interconnect diagrams.

4.2 Electromagnetic Emission

Table 4-1 Emission Declaration

EMC Emissions Guidance & Declaration for EQUIPMENT and/or SYSTEM		
The EQUIPMENT and/or SYSTEM is intended for use in the electromagnetic environment specified below. The customer or the user of the EQUIPMENT and/or SYSTEM should assure that it is used in such an environment.		
Emissions Test	Compliance	Electromagnetic Environment Guidance
RF emissions CISPR 11	Group 1	The EQUIPMENT and/or SYSTEM uses RF energy only for its internal function. Therefore, its RF emissions are very low and are not likely to cause any interference in nearby electronic equipment.
RF emissions CISPR 11	(Phase 1 - only) Class A + 12 (Phase 2 - only) Class A	(Phase 1 only) <i>Warning:</i> This equipment is allowed to be installed only in X-ray protected rooms, which provide an attenuation of at least 12 dB for radio disturbances from 30 MHz to 1 GHz. When installed in such a shielded location:
Harmonic emissions IEC 61000-3-2	Not applicable	The EQUIPMENT and/or SYSTEM is suitable for use in all establishments other than domestic and those directly connected to the public low-voltage power supply network that supplies buildings used for domestic purposes.
Voltage fluctuations/ flicker emissions IEC 61000-3-2	Not applicable	

4.3 Electromagnetic Immunity

Table 4-2 Immunity Declaration

EMC Immunity Guidance & Declaration for EQUIPMENT and/or SYSTEM			
The EQUIPMENT and/or SYSTEM is intended for use in the electromagnetic environment specified below. The customer or the user of the EQUIPMENT and/or SYSTEM should assure that it is used in such an environment.			
Immunity Test	IEC 60601-1-2 Test Level	Compliance Level	Electromagnetic Environment Guidance
Electrostatic discharge (ESD) IEC 61000-4-2	± 6 kV contact ± 8 kV air	± 6 kV contact ± 8 kV air	Floors should be wood, concrete, or ceramic tile. If floors are covered with synthetic material, the relative humidity should be at least 30%.
Electrical fast transient/burst IEC 61000-4-4	± 2 kV for power supply lines ± 1 kV for input/output lines	± 2 kV for power supply lines ± 1 kV for input/output lines	Mains power quality should be that of typical commercial or hospital environment.
Surge IEC 61000-4-5	± 1 kV line-line ± 2 kV line-earth	± 1 kV line-line ± 2 kV line-earth	Mains power quality should be that of a typical commercial or hospital environment.
Voltage dips, short interruptions and voltage variations on power supply input lines IEC 61000-4-11	< 5 % U_T (> 95% dip in U_T) for 5 sec	< 5 % U_T (> 95% dip in U_T) for 5 sec	Mains power quality should be that of a typical commercial or hospital environment. If the user of the EQUIPMENT and/or SYSTEM requires continued operation during power mains interruptions, it is recommended that the EQUIPMENT and/or SYSTEM be powered from an uninterruptible power supply or a battery.
Powerfrequency (50/60 Hz) magnetic field IEC 61000-4-8	3 A/m	3 A/m	Power frequency magnetic fields should be at levels characteristic of a typical location in a typical commercial or hospital environment.
Note : U_T is the a.c. mains voltage prior to application of the test level.			

Table 4-3 Immunity Declaration con't

EMC Immunity Guidance & Declaration for EQUIPMENT and/or SYSTEM			
The EQUIPMENT and/or SYSTEM is intended for use in the electromagnetic environment specified below. The customer or the user of the EQUIPMENT and/or SYSTEM should assure that it is used in such an environment.			
Immunity Test	IEC 60601-1-2 Test Level	Compliance Level	Electromagnetic Environment Guidance
Conducted RF IEC 61000-4-6	3 VRMS 150kHz to 80 MHz	3 V 150kHz to 80 MHz	Portable and mobile RF communications equipment should be used no closer to any part of the EQUIPMENT and/or SYSTEM, including cables, than the recommended separation distance calculated from the equation appropriate for the frequency of the transmitter. Recommended Separation Distance $d = \left[\frac{3.5}{3} \right] \sqrt{P}$ (see Table 4-4) $d = \left[\frac{3.5}{3} \right] \sqrt{P}$ 80 MHz to 800 MHz (see Table 4-4) $d = \left[\frac{7}{3} \right] \sqrt{P}$ 800 MHz to 2.5GHz (see Table 4-4) where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer and d is the recommended separation distance in metres (m). Field strengths from fixed RF transmitters, as determined by an electromagnetic site survey^a, should be less than the compliance level in each frequency range^b. Interference may occur in the vicinity of equipment marked with the following symbol: 
Radiated RF IEC 61000-4-3 (alternative method: IEC 61000-4-21)	3 V/m 80 MHz to 2.5 GHz	3 V/m 80 MHz to 2.5 GHz	
a Field strengths from fixed transmitters, such as base stations for radio (cellular/cordless) telephones and land mobile radios, amateur radio, AM and FM radio broadcast, and TV broadcast cannot be predicted theoretically with accuracy. To assess the electromagnetic environment due to fixed RF transmitters, and electromagnetic site survey should be considered. If the measured field strength in the location in which the EQUIPMENT and/or SYSTEM is used exceeds the applicable RF compliance level above, the EQUIPMENT and/or SYSTEM should be observed to verify normal operation. If abnormal performance is observed, additional measures may be necessary, such as re-orienting or relocation the EQUIPMENT and or SYSTEM. b Over the frequency range 150 kHz to 80 MHz, field strengths should be less than 3 V/m. Note : These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects, and people.			

Table 4-4 Separation Distances

Recommended separation distances between portable and mobile RF communications equipment and the EQUIPMENT and/or SYSTEM .			
The EQUIPMENT and/or SYSTEM is intended for use in an electromagnetic environment in which radiated RF disturbances are controlled. The customer or the user of the EQUIPMENT and/or SYSTEM can help prevent electromagnetic interference by maintaining a minimum distance between portable and mobile RF communications equipment (transmitters) and the EQUIPMENT and/or SYSTEM as recommended below, according to the maximum output power of the communications equipment.			
Rated Maximum Output Power (P) of Transmitter Watts (W)	Separation distance according to frequency of transmitter		
	150 kHz to 80 MHz	80 MHz to 800MHz	800 MHz to 2.5 GHz
	$d = \left[\frac{3.5}{3} \right] \sqrt{P}$	$d = \left[\frac{3.5}{3} \right] \sqrt{P}$	$d = \left[\frac{7}{3} \right] \sqrt{P}$
	Separation Distance meters	Separation Distance meters	Separation Distance meters
0.01	0.12	0.12	0.23
0.1	0.37	0.37	0.74
1	1.17	1.17	2.33
10	3.69	3.69	7.38
100	11.7	11.7	23.3
For transmitters rated at a maximum output power not listed above, the separation distance can be estimated using the equation in the corresponding column, where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer.			
Note 1: At 80 MHz and 800 MHz, the separation distance for the higher frequency range applies.			
Note 2: These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects, and people.			

4.3.1 Limitations Management :

Adhering to the distance separation recommended in [Table 4-4](#), between 150kHz & 2.5GHz, will reduce disturbances recorded at the image level but may not eliminate all disturbances. However, when installed and operated as specified herein, the system will maintain its essential performance by continuing to acquire, display, and store diagnostic quality images safely.

(*) For example, a 1W mobile phone (800MHz to 2.5GHz carrier frequency) shall be put 2.3 meters apart from the **EQUIPMENT and/or SYSTEM** (in order to avoid image interference risks).

4.4 Use Limitation :

4.4.1 External components

The use of accessories, transducers, and cables other than those specified may result in degraded ELECTROMATHNETIC COMPATIBILITY of the **EQUIPMENT and/or SYSTEM**.

4.5 Installation Requirements & Environment Control :

In order to minimize interference risks, the following requirements shall apply.

4.5.1 Cable Shielding & Grounding

All interconnect cables to peripheral devices must be shielded and properly grounded. Use of cables not properly shielded and grounded may result in the equipment causing radio frequency interference.

4.5.2 This product complies the radiated emission as per CISPR11 Group 1 Class A standard limits

The [EQUIPMENT and/or SYSTEM] is predominantly intended for use, in non-domestic environments, and not directly connected to the Public Mains Network. The [EQUIPMENT and/or SYSTEM] is predominantly intended for use (e.g. in hospitals) with a dedicated supply system, and with z X-ray shielded room. In case of using in a domestic environment (e.g. doctor's offices), in order to avoid interferences, it is recommended to use a separated AC power distribution panel & line with a X-ray shielded room.

4.5.3 Subsystem & Accessories Power Supply Distribution

All components, accessories subsystems, systems which are electrically connected to the [EQUIPMENT and/or SYSTEM], must have all AC power supplied by the same power distribution panel & line.

4.5.4 Stacked Components & Equipment

The [EQUIPMENT and/or SYSTEM] should not be used adjacent to or stacked with other equipment; if adjacent or stacked use is necessary, the [EQUIPMENT and/or SYSTEM] should be observed in order to verify normal operation in the configuration in which it will be used.

4.5.5 Low Frequency Magnetic Field

In case of a digital [EQUIPMENT and/or SYSTEM], the Gantry (digital detector) shall be apart 1meter from the generator cabinet, and 1meter apart from the analog (CRT) monitors. These distance specifications will minimize the low frequency magnetic field interference risk.

4.5.6 Static Magnetic Field Limits

In order to avoid interference on the [EQUIPMENT and/or SYSTEM] system, static field limits from the surrounding environment are specified.

Static field is specified less than <1 Gauss in Examination room, and in the Control Area.

Static field is specified less than <3 Gauss in the Technical Room.

4.5.7 Electrostatic Discharge Environment & Recommendations

In order to reduce electrostatic discharge interference, install a charge dissipative floor material to avoid electrostatic charge buildup.

The relative humidity shall be at least 30 percent.

The dissipative material shall be connected to the system ground reference, if applicable.

Chapter 2

Pre-Installation Overview

Before a LightSpeed 5.X system can be installed, all pre-installation requirements must be complete.

- Chapter 4, Sections [7.0](#) and [8.0](#) Structural Requirements
- Chapter 4, Section [9.0](#) Radiation Protection
- Broadband standard
- [Chapter 5, Environmental Conditions](#), Sections [1.0](#), [2.0](#), & [3.0](#) HVAC Requirements
- [Chapter 8, Power Requirements](#)
- [Chapter 9, Interconnection Data](#)

Section 1.0

Site Ready for Installation

Site-specific items must be verified before the installation can begin.

1.1 Dust/Dirt Contamination

The LightSpeed 5.X systems (consisting of: Console, PDU, Table and Gantry) are highly susceptible to airborne contaminants, especially concrete and drywall dust. Due to the possibility of contamination, these systems should NEVER be installed in a construction site.



NOTICE Any site with unfinished floors, walls or ceilings is considered a construction site, and is not suitable for system installation.

1.2 Chemical Contamination

Wet film processors must never be installed in the same room as the scanner, due to the possibility of chemical contamination of LightSpeed Series components. Such chemicals can contribute to increased equipment failures, increased system downtime, and decreased reliability. Film processor equipment installation must meet the manufacturer's requirements (e.g. ventilation specifications) and all applicable national and local codes. Also, consideration's should be given to the location of this equipment and chemical fumes relative to human contact as it relates to locating this equipment and chemicals in the control room.

1.3 Walls, Ceiling, and Floor

All walls, ceiling, and flooring must be completed before installation can begin.
LightSpeed 5.X scanners can only be installed on a 4" concrete floor.

1.4 Broad-band

For information on Broad-band requirements, refer to Chapter 4, [Section 8.0, on page 65](#).

1.5 Phone Line (for optional modem)

Two phone lines must be installed at or near the console and be operational prior to installation.

- 1-Analog line (for modem use)
- 1-Voice line

1.6 Review

The LightSpeed Series systems use adjustable leveling pads to support the gantry and table. The gantry has four (4) primary leveling pads. The table has five (5) pads used for leveling it.

Using the GE print to establish the room layout, make sure all the operating and service clearances shown on the print are observed. Using the template (P/N 2363851) shipped with the system, locate the anchor holes. Make sure they clear structural interferences in the floor.

Clean the area. Free the mounting surface of any material that may interfere with the positioning and leveling of the system.

- 1.) Lay out the 2 floor templates.
- 2.) Start with the Gantry template—align per the GE print.
- 3.) Place the table template over the top of the Gantry template. Align the scan and table center-lines and secure the templates to the floor. Make sure there are no potential clearance issues.
- 4.) Check the level of the floor (See [Figure 2-1](#)) across the templates.

Note: Tiles (or other resilient flooring) around all holes will be cut during the installation process.

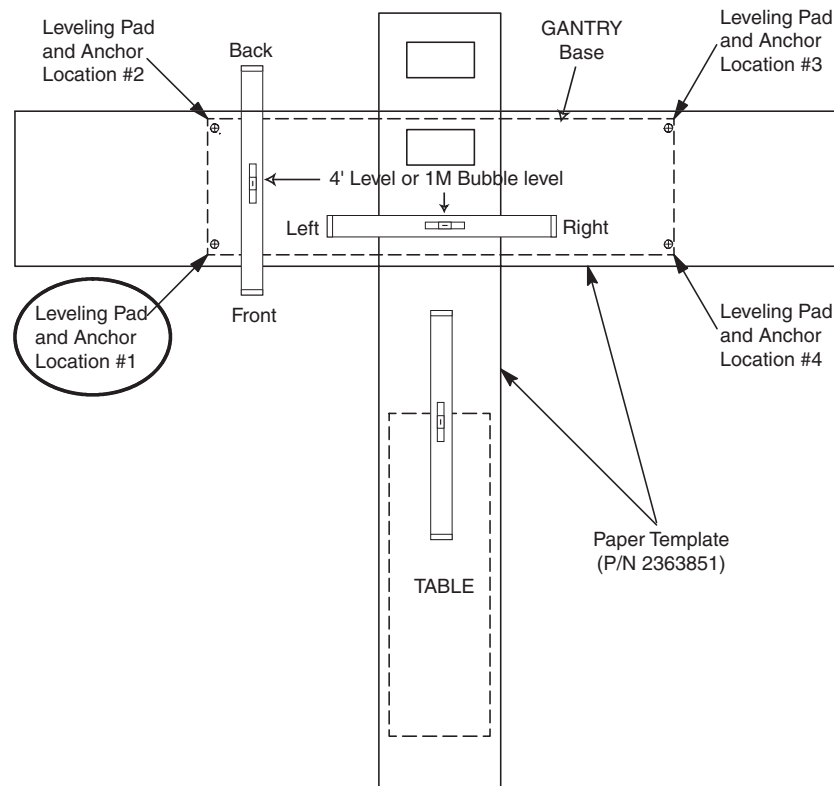


Figure 2-1 Hole Locations

Section 2.0

Pre-Installation Checklist

Required Information for Site

Must be completed before the scheduled delivery date

Hospital Name as it appears on the system screens: _____

Network ID numbers / IP addresses

Camera: _____

PACS: _____

AW: _____

Other - Specify type & ID: _____

Other - Specify type & ID: _____

Camera setup information: _____

AW Direct Connect address: _____

Do you want HIPAA enabled? No___ Yes___

Do you want automatic downloads enabled? No___ Yes___

:

Table 2-1 Schedule Date Commitments

GE		CUST		Dates
Y	N	Y	N	
		☐	☐	Has the project schedule been verified with facilities department, contractor, and GE?
		☐	☐	Will the committed site-ready date be met?
☐	☐	☐	☐	Does the completion date for any/all construction meet or precede the delivery date?
☐	☐	☐	☐	Is the Power & Ground survey complete? Date: _____
				Hospital contact: _____
		☐	☐	Site-Ready visit is scheduled. Date: _____
		☐	☐	Delivery date is scheduled. Date: _____
		☐	☐	Installation date is scheduled. Date: _____
☐	☐	☐	☐	Installation timing: A: Weekdays___ B: Weekend___ C: Quick Install___
				If B or C, have all sub-contractors been notified? No___ Yes___
☐	☐	☐	☐	Does the delivery and/or installation date need to be adjusted?
		☐	☐	First-Use date is scheduled. Date: _____
☐	☐	☐	☐	Applications/Training dates: On-Site Training Date: _____
				Healthcare Institute Training Date: _____

Table 2-2 General Site Planning

GE		CUST		General / Site Requirements
Y	N	Y	N	<i>Must be completed 5 weeks before scheduled delivery date</i>
☐	☐	☐	☐	Have final drawings been approved and distributed to the contractors?
☐	☐	☐	☐	Are final drawings "signed off" to approve equipment layout / orientation?
☐	☐	☐	☐	Do the actual room dimensions match those on the final drawings?
		☐	☐	Has the radiologist health physician reviewed and approved the room layout and shielding requirements?
		☐	☐	Have any additional requirements or questions about the installation been discussed with GE? List: _____ _____ _____ _____
		☐	☐	Is there a person assigned to review and verify that all installation requirements are met? Name: _____
		☐	☐	Have the specific site requirements been discussed with the contractor? Refer to the GE final drawings specifications. (See Table 2-3 , below)
		☐	☐	Has the responsibility of cabling, installing, and interfacing accessories not on the order been discussed?
		☐	☐	Are all third-party vendors identified, notified and scheduled? (Examples: Netcom, Medrad, etc.)
		☐	☐	Have all regulatory requirements been met per Section 2.0, on page 42
		☐	☐	Will existing network, broadband, and camera cable drops reach new locations and will they meet the requirements and function with LightSpeed 5.X 16, Ultra & Plus ? If not, what are the requirements? List: _____ _____ _____

Table 2-3 References for Specific Site Requirements

Sections for Specific Requirements	
Room Planning - Chapter 4, Room Planning	Floor Loads & Weights - Chapter 6, Floor Loading and Weights
Radiation Protection - Section 9.0, on page 69	Delivery - Chapter 7, Delivery Data
Environment - Chapter 5, Environmental Conditions	Power - Chapter 8, Power Requirements
<u>All work</u> by contractors must be completed before the scheduled delivery date.	

Table 2-4 Equipment Compatibility

GE	CUST	Equipment
Y N	Y N	<i>Must be completed 5 weeks before scheduled delivery date</i>
☐ ☐	☐ ☐	Has the order been reviewed for completeness and compatibility with existing equipment? Typical equipment: Remote monitors ____ AW relocation ____ Cardiac option ____ Injectors ____
☐ ☐		Are interfaces to existing and/or new accessories ordered and planned for accordingly?
☐ ☐	☐ ☐	Have the following peripheral locations been included in the site drawings? EKG monitor ____ Injector control ____ Laser camera ____ UPS ____ 2 nd Monitor ____
☐ ☐	☐ ☐	Will GE Healthcare provide additional services per contract negotiations?
☐ ☐		Are correct length cables on order?

Table 2-5 Network Connections

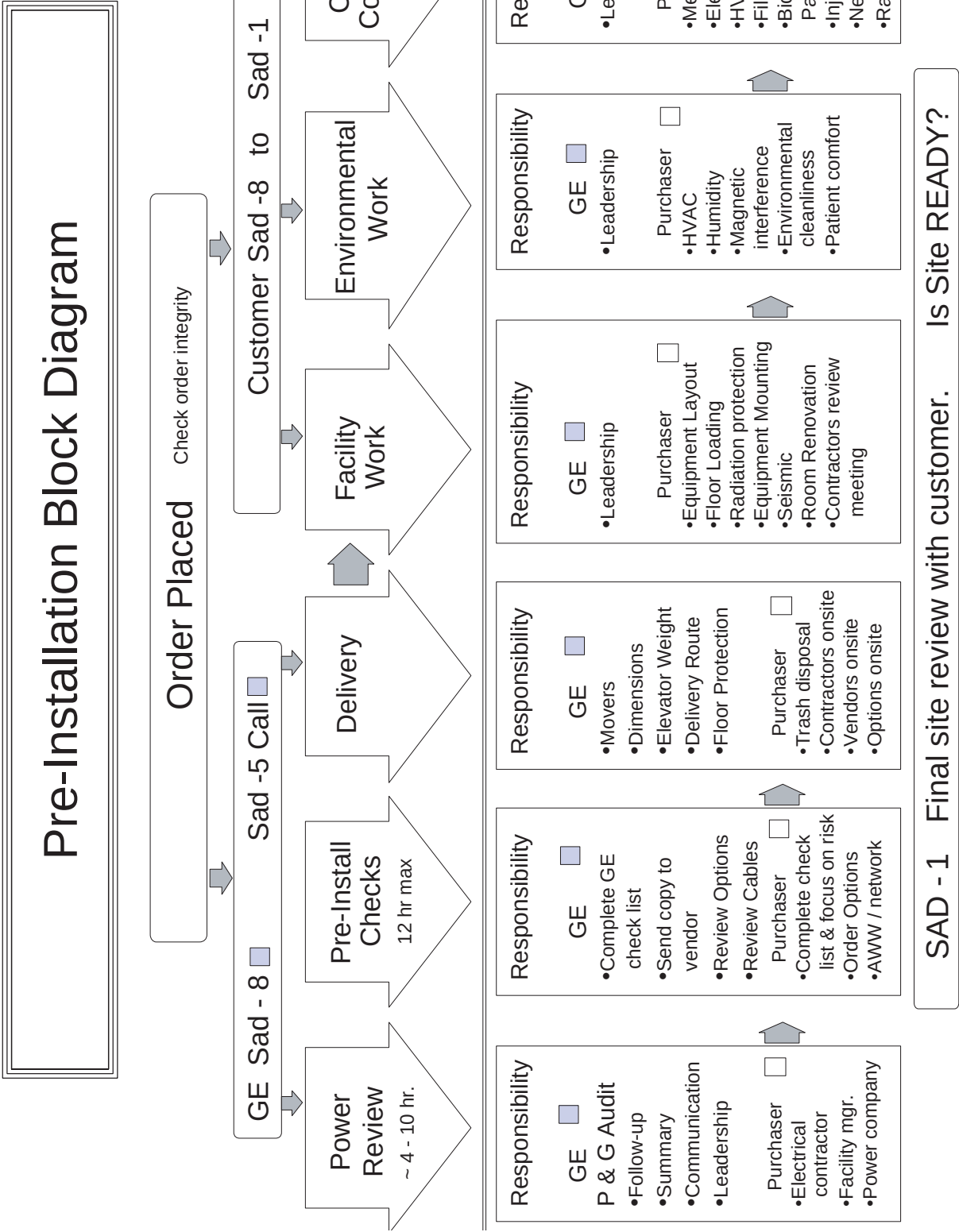
GE	CUST	Network Installation and Setup
Y N	Y N	<i>Must be completed 5 weeks before scheduled delivery date</i>
☐ ☐	☐ ☐	Have IP addresses and Host Names been obtained? No ____ Yes ____
☐ ☐	☐ ☐	Will a network camera be used? No ____ Yes ____
☐ ☐	☐ ☐	Mandatory: Is the network installed, are network jacks installed, and is the entire network tested?
☐ ☐	☐ ☐	Mandatory: Broadband VPN installed/setup?
☐ ☐	☐ ☐	Mandatory: Are network software options ordered ____ HIS RIS option ____ DICOM print ____ AW ____
	☐ ☐	Optional: Has modem option ordered? ____ (Requires a site escalation)
☐ ☐	☐ ☐	Optional: Is the LightSpeed 5.X Pro16 service telephone line identified and installed for InSite? (Electrical, mechanical, etc.)

Table 2-6 Miscellaneous Tasks

GE	CUST	Other
Y N	Y N	<i>Must be completed before the scheduled delivery date</i>
	☐ ☐	Arrangements made in the schedule to allow adequate time for remodeling, if required (such as wall, floor, or ceiling repair work, painting, other cosmetic finishes)
	☐ ☐	Have arrangements been made to clean the floor <i>after</i> equipment removal and <i>prior</i> to the installation of the new equipment?
☐ ☐	☐ ☐	Is de-installation of existing equipment required? No ____ Yes ____ Removal date ____
☐ ☐	☐ ☐	Is there a trade-in of existing equipment? No ____ Yes ____ GoldSeal ____

Table 2-6 Miscellaneous Tasks (Continued)

GE		CUST		Other
Y	N	Y	N	<i>Must be completed before the scheduled delivery date</i>
☐	☐	☐	☐	Delivery route identified and verified with the proper hospital personnel? No__ Yes __
☐	☐	☐	☐	Elevators and doors checked for size and weight constraints? No__ Yes __
☐	☐	☐	☐	Have appropriate arrangements been made with traffic for delivery?No__ Yes __
☐	☐	☐	☐	Will acceptance/performance testing or bio-medical testing be required?No__
				Yes __ Date: _____
☐	☐	☐	☐	Are trash and/or recycling bins available for the removal of papers, boxes, etc. during the installation? No__ Yes __



Rev 9

Chapter 3

System Catalog

Section 1.0

Option Catalog Numbers

The following is a list of system options requiring site planning work for the LightSpeed 5.0 system. Contact your local GE Medical Systems Sales representative for a complete list of all system options or visit us at <http://www.gemedical.com>. Refer to the instruction manuals supplied with specific options for respective details.

CATALOG NUMBER	OPTION DESCRIPTION
International Dolly Set B7850LD or B78502LD	For International customers, if dollies are required.
Remote Color Monitors B7858LC or B78582LC B7530LC	LightSpeed 5.0 LCD monitor. Remote 20 inch diagonal color monitor.
SmartScore Option B7850PL	EKG Monitor and Recording Device
Connect Pro Option B7500LN	ConnectPro Option provides a direct interface to HIS/RIS
Bar Code Reader B7540RB	LightSpeed 5.0 Bar Code Reader
Long Cable Set B7816PH	
Modem B7700MG	Global Modem Kit

Table 3-1 System Options Catalog (Part) Numbers

Section 2.0

Base Scanner System

2.1 Application

The CT scanner system includes hardware and software to support patient data acquisition and image analysis for whole-body computed tomography.

2.2 Configuration

The base scanner system is configured as shown. All scan and analysis functions are controlled from the operator's console (not shown).



Figure 3-1 Base Scanner System

Chapter 4

Room Planning

Section 1.0

Required Systems Clearances



NOTICE If your system was installed before December 2006, Limited Access is not available for your system. You must use the Preferred Room Size to reinstall it.

Consult your local GE Sales and Service Representative about your specific needs.

Some possible room size dimensions are shown in the table below. These room size dimensions are table dependent.

Limited Access Rooms (width only):

Minimum Width:	3556 mm (356 mm cover to wall) 11 ft. 8 in. (14 in. cover to wall)
Average Width:	3912 mm (711 mm cover to wall) 12 ft. 10 in. (28 in. cover to wall)

Additional component dimensions are available in [Figure 4-11](#) through [Figure 4-14](#) of this document. Consult your local General Electric Project Manager of Installation (GE PMI) for your appropriate room specifications.

For equipment clearance requirements, refer to [Section 2.0](#). Remember, sufficient Regulatory and Service clearances must be maintained around equipment for full operation, service and safety.

Cable length is an important consideration in room layout. The LightSpeed 5.X system is shipped with standard short length cables, with a set of longer cables (B7816PH) available as an option. See the electrical page of the GE print for your specific requirements.

Note also, that where possible, the cables should enter the gantry from the rear, utilizing the rear cable cover assembly. Alternate cable entry is possible at the center of the gantry (refer to the Installation template).

- Excess cable length cannot be stored behind the operator console or PDU.
- A long cable must not be cut or shortened.
- Excess cable may be stored in the cable wall or floor duct, provided sufficient space is available.
- All NEC 70-E Electrical Regulations regarding conduit or duct fill must be observed.

Note: See [Section 4.0](#) for Service Clearances.

2.1.1 Regulated Minimum Working Clearance by Major Subsystem

- Requirements apply to equipment operating at 600 V or less, where examination, adjustment, servicing, or maintenance is likely to be performed while live parts are exposed.
- Direction of Service Access is defined as perpendicular to the surface of the equipment being serviced.
- Required regulatory clearance distances must be maintained and may not be used for storage. This includes normal system operation as well as service inspection or maintenance.

Work Space Requirement	Minimum Clear Space in Inches	Additional Conditions
Direction of Service Access	N/A No exposed live part hazards	

Table 4-1 Console

Work Space Requirement	Minimum Clear Space in Inches	Additional Conditions
Direction of Service Access (Front of PDU)	914.4 mm (36 in.)	1219.2 mm (48 in.), if exposed live parts of 151 - 600 volts are present on both sides of workspace with the operator between. 1066.8 mm (42 in.), if opposite wall is grounded and exposed live parts of 151 - 600 volts are present.
Service Access Width (Left-Right of workspace)	762 mm (30 in.)	This is the width of the working space in front of the equipment. 30 inches (762 mm) min or the width of the equipment, whichever is greater
Head Clearance	1981.2 mm (78 in.)	The height of the workspace measured from floor at the front edge of equipment to ceiling or overhead obstruction(s). 78 inches (1981.2 mm) or height of equipment, whichever is greater

Table 4-2 NGPDU

Work Space Requirement	Minimum Clear Space in Inches	Additional Conditions
Direction of Service Access (All Sides)	914.4 mm (36 in.)	1219.2 mm (48 in.), if exposed live parts of 151 - 600 volts are present on both sides of workspace with the operator between. 1066.8 mm (42 in.), if opposite wall is grounded and exposed live parts of 151 - 600 volts are present.
Service Access Width (Left-Right of workspace)	762 mm (30 in.)	This is the width of the working space in front of the equipment. 762 mm (30 in.) min or the width of the equipment, whichever is greater.

Table 4-3 Gantry

Work Space Requirement	Minimum Clear Space in Inches	Additional Conditions
Direction of Service Access (Table Head)	N/A	1219.2 mm (48 in.), if exposed live parts of 151 - 600 volts are present on both sides of workspace with the operator between. 1066.8 mm (42 in.), if opposite wall is grounded and exposed live parts of 151 - 600 volts are present.
Direction of Service Access (Table Sides)	914.4 mm (36 in.)	*Can be reduced to 711.2 mm (28 in.) provided a written and signed approval is obtained by the local team from the local AHJ (Authority Having Jurisdiction). Signed document must be on file within GE.
Direction of Service Access (Table Foot)	711.2 mm (28 in.)	457.2 mm (18 in.) minimum for Front Gantry Cover removal only if unobstructed egress space of 711.2 mm (28 in.) is maintained around the equipment for room exit. This also means no trip hazards exist along the path of egress.
Service Access Width (Left-Right of workspace)	762 mm (30 in.)	This is the width of the working space in front of the equipment. 762 mm (30 in.) min or the width of the equipment, whichever is greater.

Table 4-4 Table

2.1.2 Terms and Definitions

EGRESS

The path of exit from within any room. U.S. regulatory requires a minimum of 711.2 mm (28 in.) of continuous and unobstructed space including trip hazards along the path of exit.

WORK SPACE

This is the dimensional box required for safe inspection or service of energized equipment. It consists of depth, width, and height. The depth dimension is measured perpendicular to the direction of access. U.S regulation is minimum of 914 mm (36 in.). Additional conditions can increase the minimum requirement. FCT defines this as the envelope of the component increase the minimum requirement. FCT defines this as the envelope of the component superstructure. For the NGPDU it is with the front panel removed. For the gantry and table, it is with the patient or external covers removed.

SERVICE ACCESS WIDTH

This is the width of the working space in front of the equipment, a minimum of 762 mm (30 in.), or the width of the equipment, whichever is greater.

HEAD CLEARANCE

This is the height dimension of "Work Space". The height of the workspace measured from floor at the front edge of equipment to ceiling or overhead obstruction(s), 1981 mm (78 in.) or height of equipment, which ever is greater.

GROUNDING WALL

Any wall that can be electrically conductive to earth ground. Masonry, concrete, or tile, are considered conductive. Additional commonly found aspects of a wall should also be considered as grounded. This is not an all-inclusive list:

- Medical Gas ports
- Metal door and window frames
- Water sources and metallic sink structures
- Metallic wall-mounted cabinets
- A1 disconnect panel
- Equipment Emergency Off panels
- Industrial equipment, such as air conditioners and vents
- Expansion joints

The following are not considered as grounded elements of a common wall:

- Standard wall outlet
- Light switches
- Telephones
- Communication wall jacks

MINIMUM

The lowest limit permitted by law or other authority.

DIMENSIONS AND CLEARANCES

Consisting of, or representing, the lowest possible amount of degree for freedom permissible for equipment siting. This relationship must meet all safety, service, and regulatory requirements to be acceptable.

PRE-INSTALLATION ESCALATION

Process to consult with CT Engineering, the Design Center or EHS regarding pre-installation issues related to your siting concerns.

Section 3.0

Additional Regulatory Clearance Information

3.1 Minimum Room Size (Limited Access)

The CT Gantry Left Side Limited Access Initiative provides the capability to reduce the minimum room size for CT Systems while still meeting all installation requirements and specifications. This adds left side flexibility, allowing the CT system to be sited in rooms with widths 558.8 mm (22 in.) smaller than the current minimum room width. Left-side access and egress may be restricted. Refer to your site's installation print for your room's detail.

Wall duct and conduit on walls within the regulatory clearance areas shall be 1067 mm (42 in.), measured from the covers to the obstruction. Servicing of the CT System can be safely performed within the regulatory envelopes, however sufficient space must be maintained to remove system covers, and replace large system components. To achieve this clearance for the gantry, clear space must be available to maneuver the gantry covers mounted on the service dollies. Surface floor raceway cannot be used in the egress route areas. OSHA ramps are available. The FE lifting the rear or front cover to avoid floor obstructions is not an EHS-approved service procedure.

One Service Engineer shall be able to accomplish all service component replace tasks listed without the need for special tools or equipment, such as a tube change, detector change, and HV tank.

3.1.1 Regulatory Caution

Site prints are required for all system installations including relocation and moves. CT room layout as shown on your site print shall meet all regulatory requirements as described in the installation manual. Additional room components such as cabinets reduce room size. Equipment not shown on the site print may void the caution statement, making the room non-compliant. Actual site measurements before installation will be taken to determine room size and compliance.

3.1.2 Egress Clearance

Egress requires a clear, unobstructed route out of the room, either around the back of the gantry or around the back of the table. If your egress route is not around the back of the table, maintain 457 mm (18 in.) of clearance between the back of the table, with a continuous width of 3200 mm (126 in.), 1600 mm (63 in.) on each side of the table center line, on each side to any obstruction so that the front cover can be removed. Refer to the Pre-Installation manual for more details on service clearances.

Exceptions

Rooms smaller than 3556 mm x 5867.4 mm (11 ft. 8 in. x 19 ft. 3 in.) require construction to meet the minimum requirements. The design center or your GE PMI may have additional recommendations for your room size.

3.1.3 Operational Caution

In a minimum room layout 355.6 mm - 685.8 mm (14 in. - 27 in.), the customer should consider workflow, customer access for patient care, and critical-care operations space requirements. Additionally, there may be limited equipment access on the gantry left side when loading patients or when positioning patient equipment in the room between the gantry and the wall. Detailed customer installation tasks are detailed in the product Pre-Installation manual, Chapters 1-4.

3.1.4 System Specifications (LS16, Ultra & Plus)

System Configuration	Preferred Room Size	Approved Room Size	Minimum Room Size
LS16, Ultra & Plus	4420 x 7772 mm ¹ (14 ft. 6 in. x 25 ft. 6 in.)	4115 mm x 5867 mm ¹ (13 ft. 6 in. x 19 ft. 3 in.)	3556 mm x 5867 mm ² (11 ft. 8 in. x 19 ft. 3 in.)

Table 4-5 System Specifications (LS16, Ultra & Plus)

¹Same Regulatory requirements apply

²Same Regulatory requirements apply, with the addition of no energized left side service.

3.1.4.1 Preferred Room Size

The "preferred room size" configuration offers the most flexibility for future upgrades. It has sufficient workspace and space to add millwork, while meeting all regulatory requirements. This room is compatible with most two-step future installations.

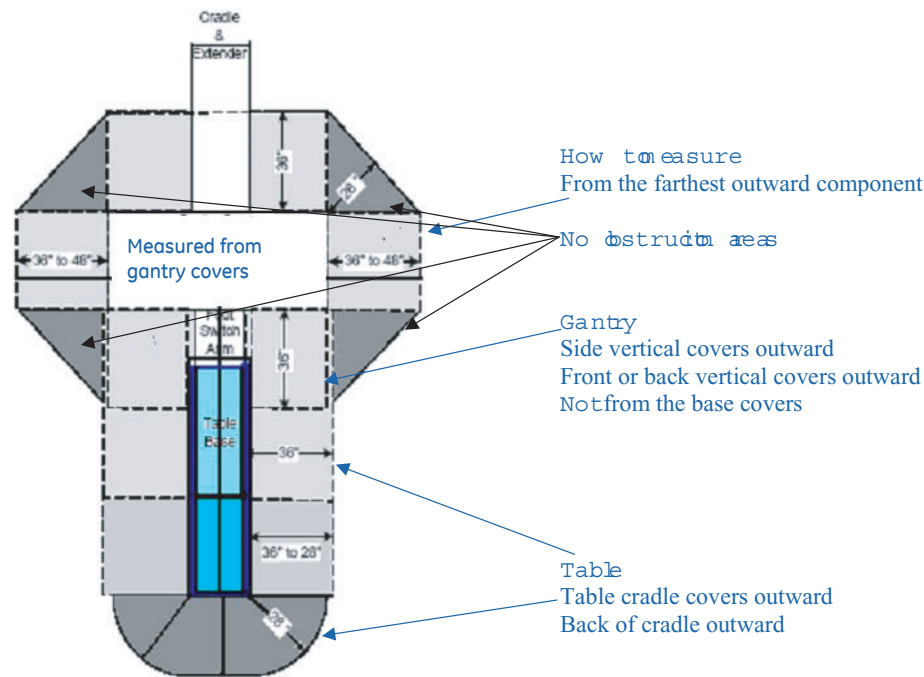
3.1.4.2 Approved Room Size

The "approved room size" configuration allows for some future upgrades. It has sufficient workspace, but limited space to add millwork and meet all regulatory requirements. This room may be compatible with some two-step future installations.

3.1.4.3 Minimum Room Size

The "minimum room size" configuration allows for no future upgrades. It has limited workspace and no in-room millwork, but meets all regulatory requirements. This room is not compatible with twostep future installations.

3.2 How to Measure

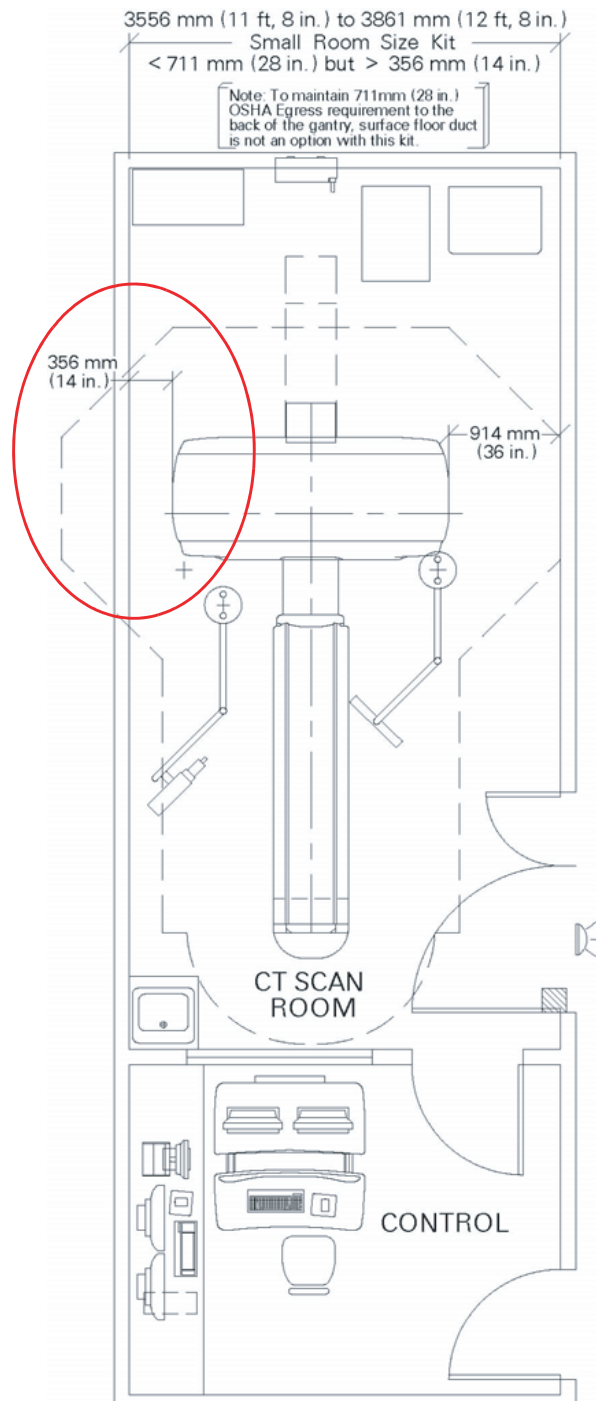


3.3 Minimum Room Size & Requirement Layouts

All gantry covers shall be removed by one person without lifting. Allow sufficient space to remove the front and back covers so that they can be "parked" during equipment service. In the limited access (Room A) configuration, carefully review user operational compatibility and placement of emergency equipment within the room.

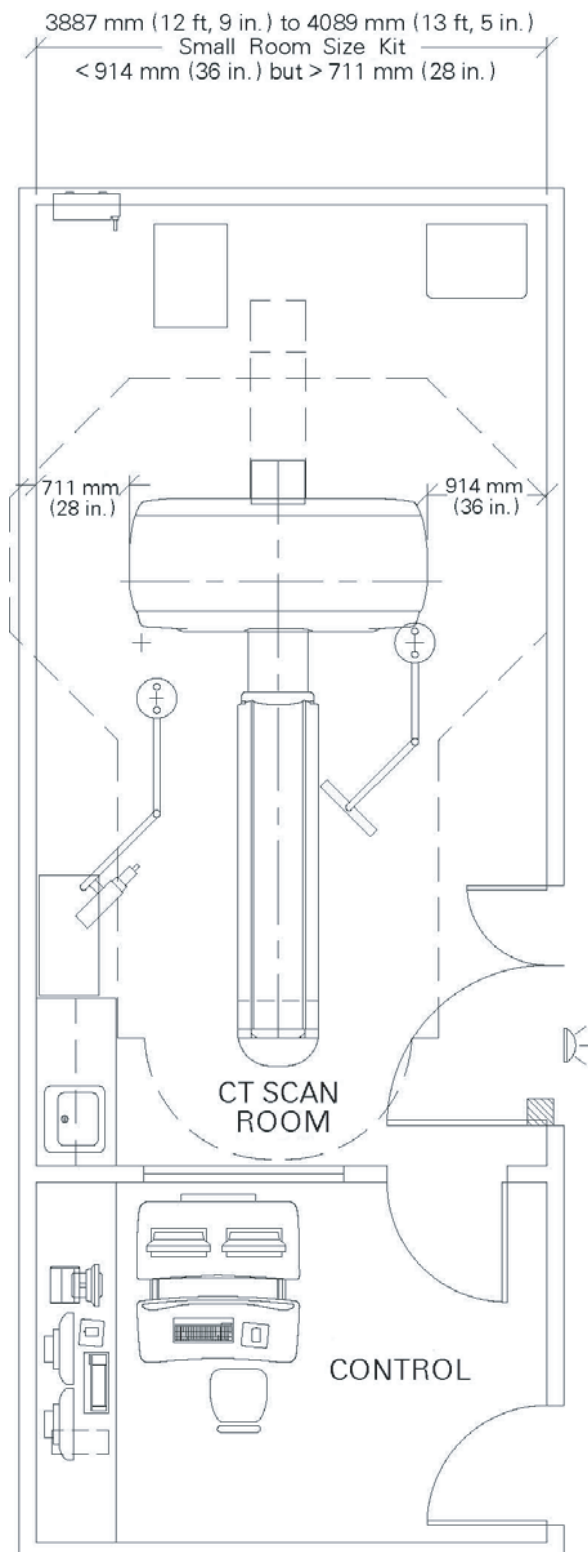
Room A - Less than 711 mm (28 in.), but greater than 256 mm (14 in.), measured from the covers to the left sidewall.

In this configuration service, egress and workspace are compromised around the gantry's left side.



Room B - Less than 914 mm (36 in.) but greater than 711 mm (28 in.) measured from the covers to the left sidewall.

In this configuration service, egress, and workspace are acceptable around the gantry.



3.4 Preferred and Approved Room Size & Requirement Layouts

Figure 4-2 and Figure 4-3 show preferred and approved room layouts, with and without in-room cabinets. You need to be aware of locations for med gas, surface duct-work, or other items that make a grounded wall.

Note: Your room layout may meet the preferred or approved room requirements but look different than Figure 4-2 or Figure 4-3. Equipment shown in these details may not represent the product you are siting. Contact your sales person to have a detail room layout completed for your site.

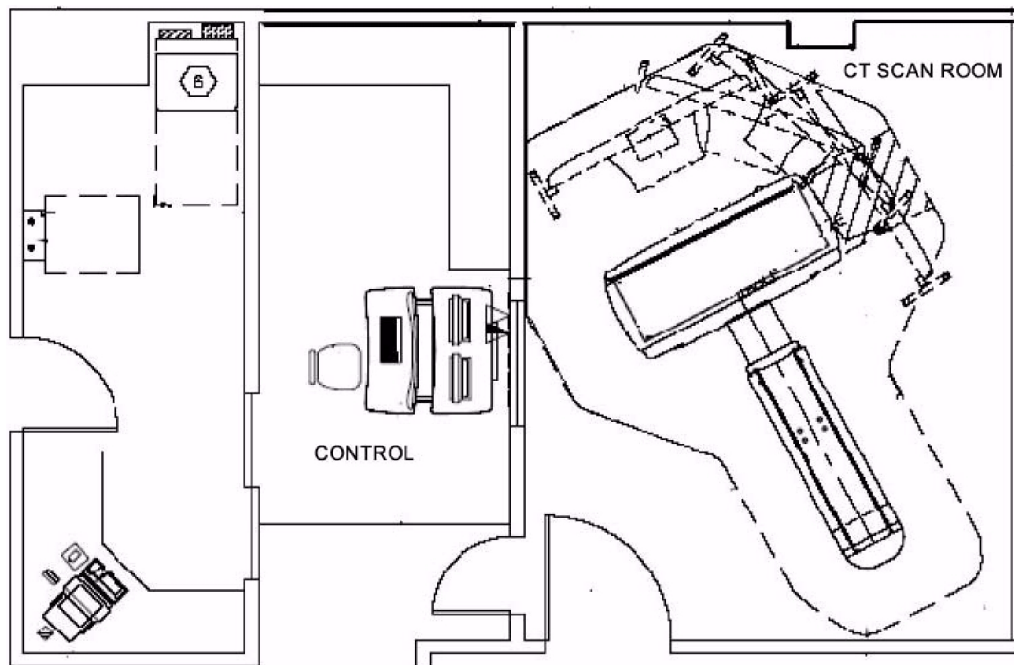


Figure 4-2 Preferred Room Layout

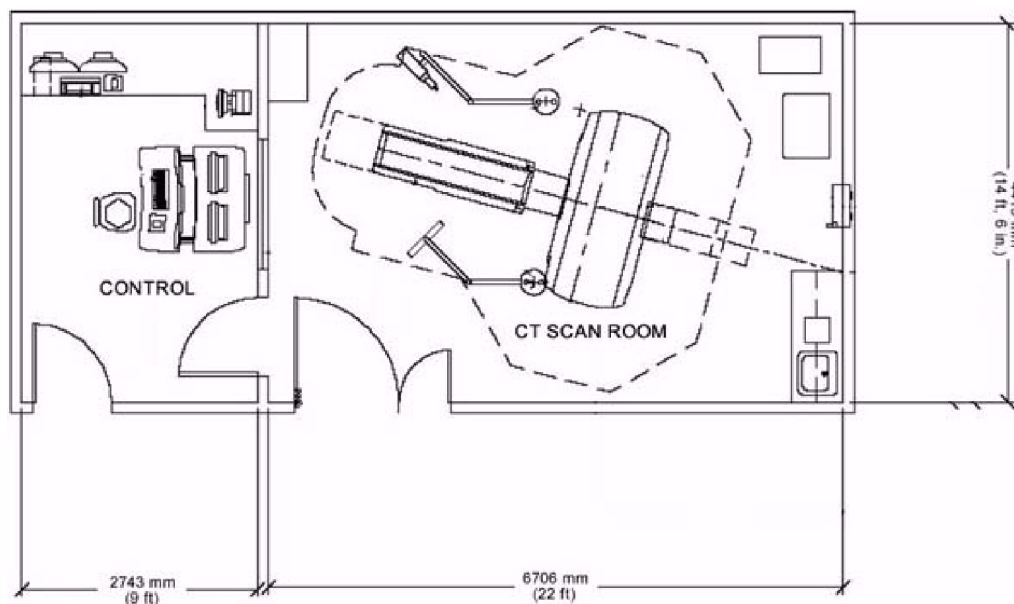


Figure 4-3 Approved Room Rayout

Section 4.0 Service Clearances

Servicing the CT System can be safely performed within the regulatory envelopes defined in [Section 2.1](#), however sufficient space must be maintained to remove the covers from the system. To achieve this clearance for the gantry, clear space must be available to maneuver the gantry covers mounted on the service dollies. One Service Engineer can accomplish this. If such space is not available, then two Service Engineers are required for gantry cover handling. This must be clearly communicated with the Customer.

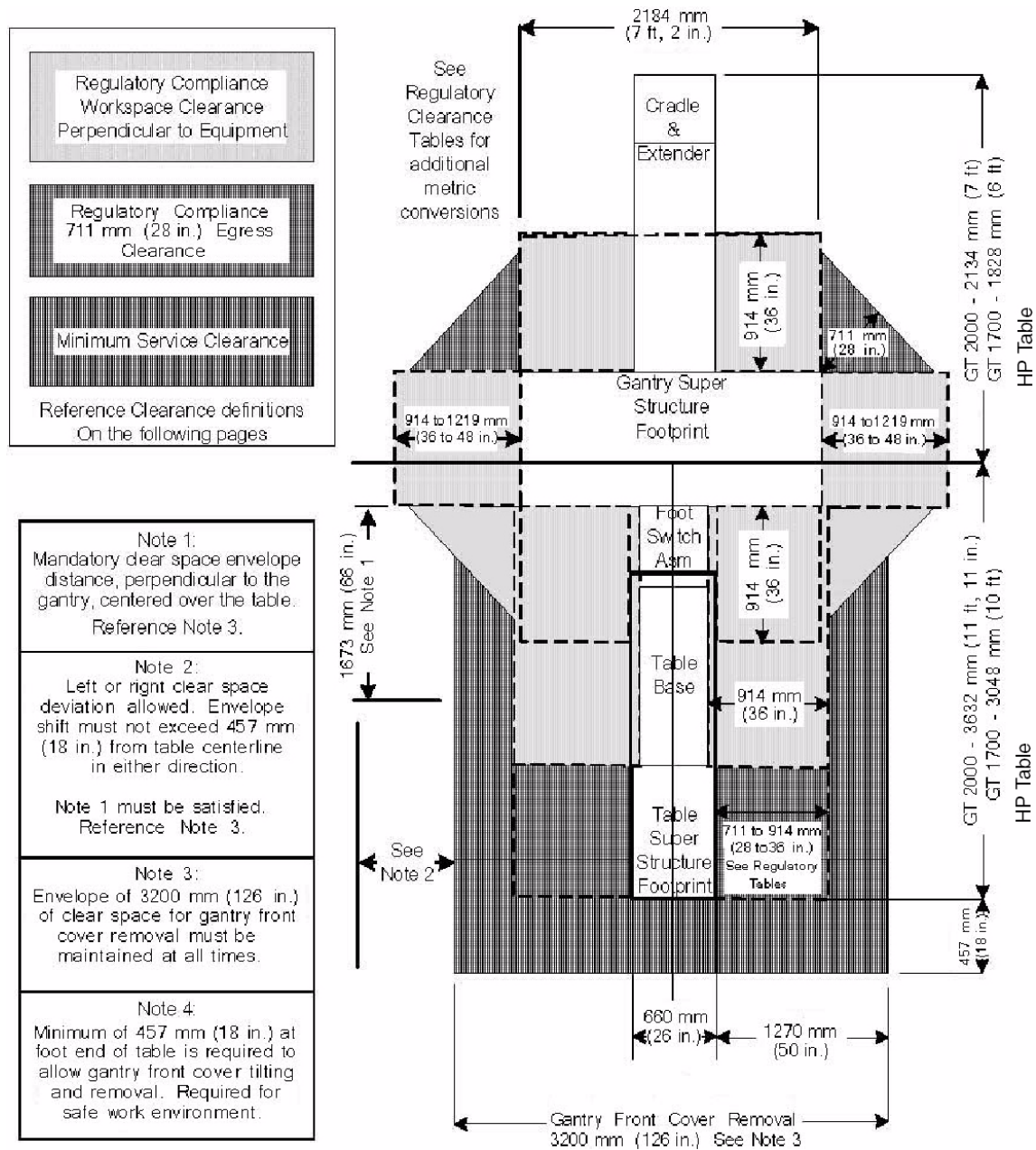


Figure 4-4 Minimum Service Clearance

4.1 Service Clearances for Single Service Engineer

- Gantry front cover removal requires the use of the Tilting Cover Dollies. These dollies allow the Service Engineer to separate the cover from the gantry, tilt the cover 90 degrees, roll the cover to the foot end of the table, and then tilt the cover an additional 90 degrees, such that the front cover is now upside down relative to the normal system-mounted condition. [Figure 4-4](#) illustrates the minimum clear space required to achieve this operation of 3,200.4 mm (126 in.). The gantry front cover must be removed to a position that satisfies the minimum regulatory clearances.
- The gantry rear cover with service dollies installed, requires a width of 2387.6 mm (94 in.) and a depth of 584.2 mm (23 in.) of clearance for removal as shown in [Section 6.0](#), [Figure 4-9](#). Sufficient space must be calculated to move the cover either straight back or to one side of the table to satisfy the minimum service clearance shown in [Figure 4-4](#). This means the rear cover cannot violate the workspace on the rear, or on either side of the gantry.
- If gantry service requires both the front and rear covers be removed, then these covers must be positioned within the room in such a manner as to not violate the egress clearances on any side of the gantry. This may necessitate removing the covers from within the suite. This should be discussed with the customer and provisions made to accommodate this potential event.
- A single Service Engineer can safely perform servicing of the table. Sufficient clear space must be available to maintain egress clearances when the table covers or cradle are removed.

4.2 Power Distribution Unit (NGPDU) Service Clearance

Positioning of this component must be considered for regulatory compliance as defined in [Section 2.1](#), Regulated Minimum Working Clearance by Major Subsystem. See Regulatory Tables.

4.3 Console Service Clearance

The operator console does not present an exposed live parts hazard. However, a minimum working space depth of 1219.2 mm (48 in.) and full width of the operator console shall be maintained at all times for service activity. Additionally, sufficient space needs to be provided for repositioning of the operator console, and side clearance for rear service access. See [Figure 4-8](#) for a typical control room layout.

Section 5.0 Recommended Layouts

Consult your local GE Sales and Service Representative about your specific needs.

Figure 4-5, Figure 4-6 and Figure 4-7 show three of many possible room layouts. For exact equipment sizes, refer to Section 6.0. Remember, sufficient space must be maintained around equipment for full operation, service and safety.

Note:
Other room
arrangements
are possible.

Cable length is an important consideration in room layout. LightSpeed 5.X system is shipped with standard length cables, with a set of longer cables (B7816PH) available as an option. See the electrical page of the GE print for your specific requirements. (The layout shown in Figure 4-6 is for standard length cables.) Note, also, that where possible, the cables should enter the gantry from below.

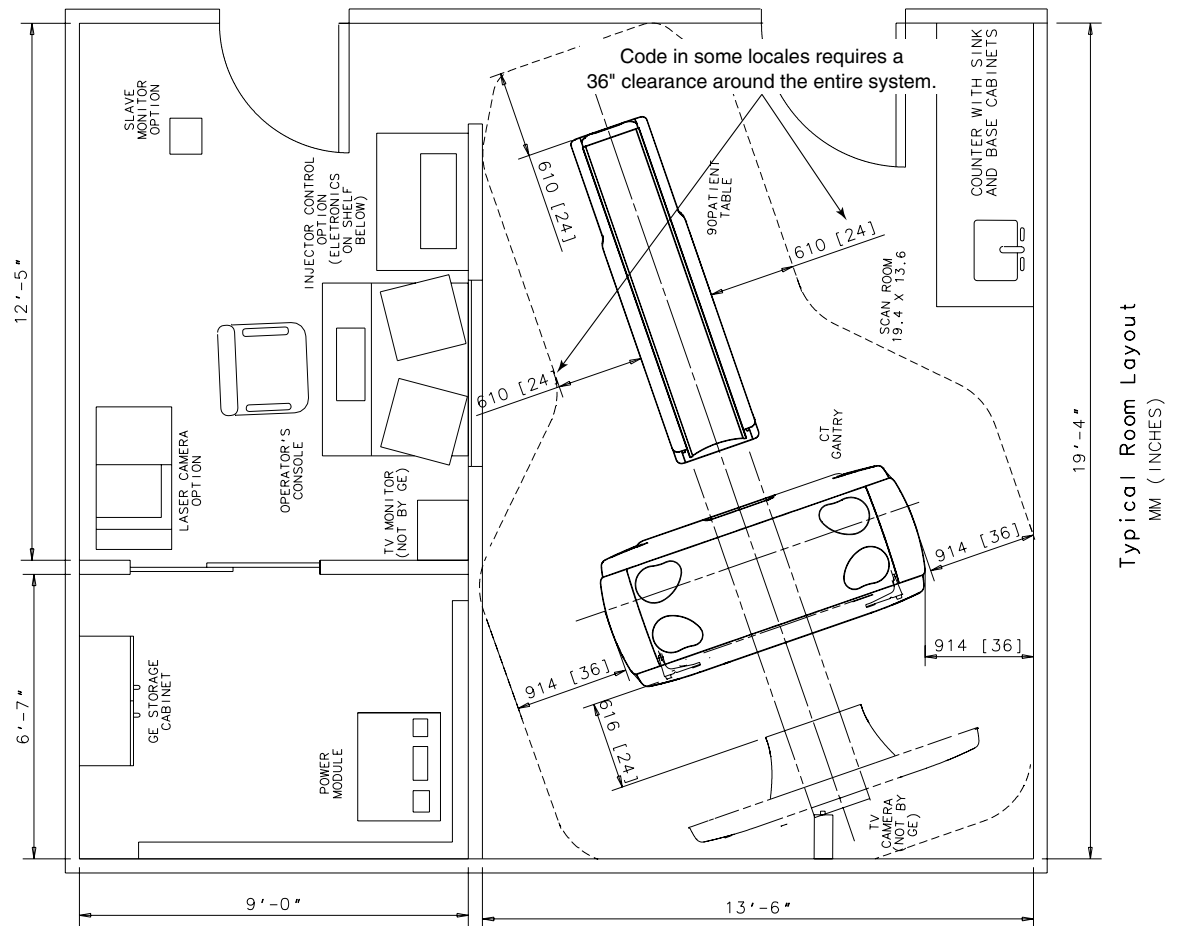


Figure 4-5 Typical Room Layout (with minimum table & gantry service clearances)

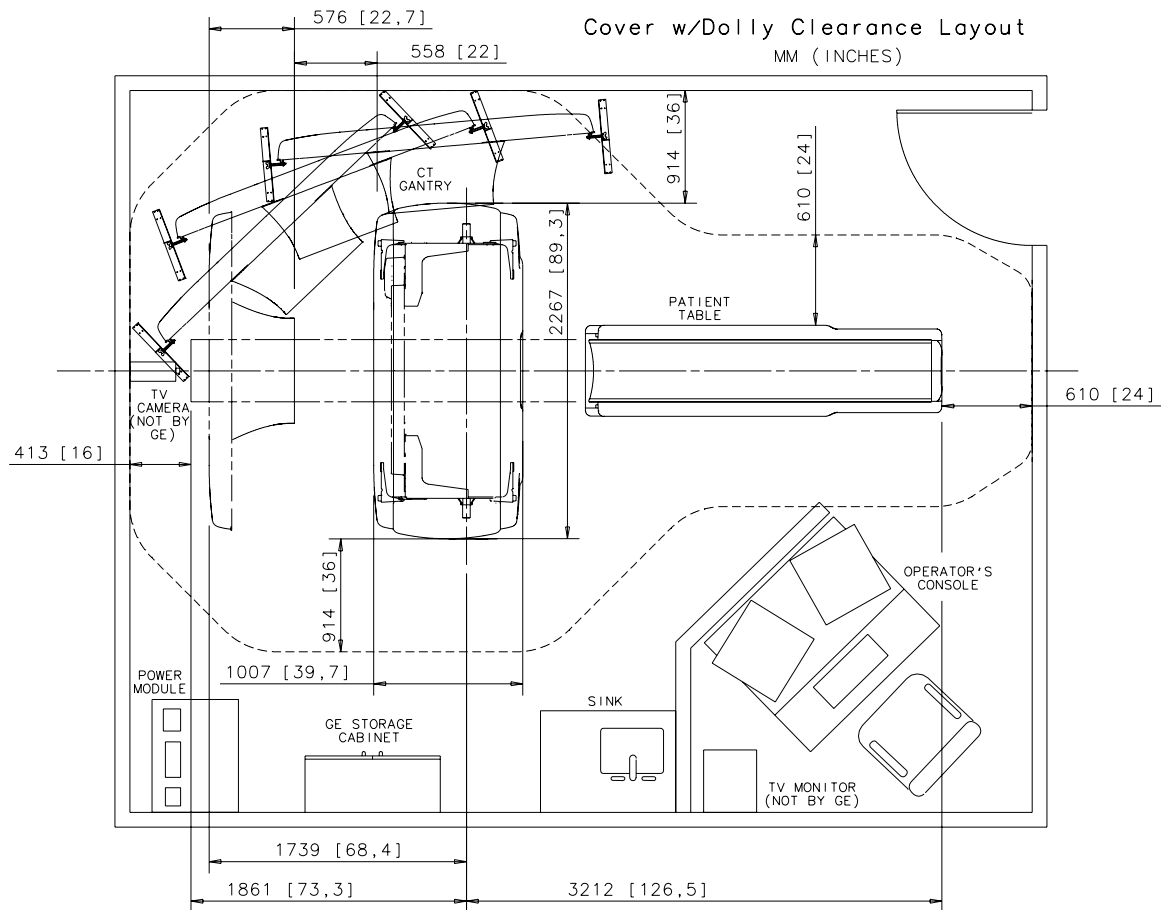


Figure 4-6 Sample Room Layout, Cover Removal Clearances

COMMON DIMENSIONS AND CLEARANCES

System Service

- Back cover removal (rear clearance) 36.0" (913.5mm)
- Front/Back cover removal (side clearance) 36.0" (913.5mm)
- Gantry service side to obstruction 36.0" (913.5mm)
- Area around the PDU 42.0" (1067mm)
- Area around the A1 breaker box 42.0" (1067mm)
- Area between the table down foot end to obstruction 36.0" (913.5mm)

Options

Ceiling Pedestal mount lowest point to floor Injector or Monitor 96.0" (2438.5mm)

System Operation

- Finished ceiling to floor 108.0" (2743mm)
- Table max extension head end with extender from Center Line 80.0" (2030.0mm)
- Table extension head end with extender to obstruction 6.0" (152.0mm)
- Table in lowest position w/cradle at home position to Center Line 126.5" (3209.0mm)
- Back of Console to wall 6.0" (152.0mm)
- Back of PDU to wall 6.0" (152.0mm)

Typical Room Dimensions

- Scan Room 13.6' X 22.0' (4.1m X 6.60m)
- Control Room 9.0' X 22.0' (2.74m X 6.60m)

Additional dimensions are available in [Figure 4-10](#) through [Figure 4-14](#) of this document. Consult your local General Electric Installation Specialist for your appropriate room specifications.

5.1 Injector Control

A suitable work area, which is within reach of the operator's console, should be provided for placement of the injector control. Refer to [Figure 4-7](#).

Wall mounted, ceiling mounted and pedestal units need cables to be routed from the gantry area to the console area. The supplied cable is 50 feet in length. Injectors require AC power. Mounts are available in different configurations and lengths. Refer to Injector documentation for detailed installation instructions. Refer to [Figure 4-7](#) for injector placement information.

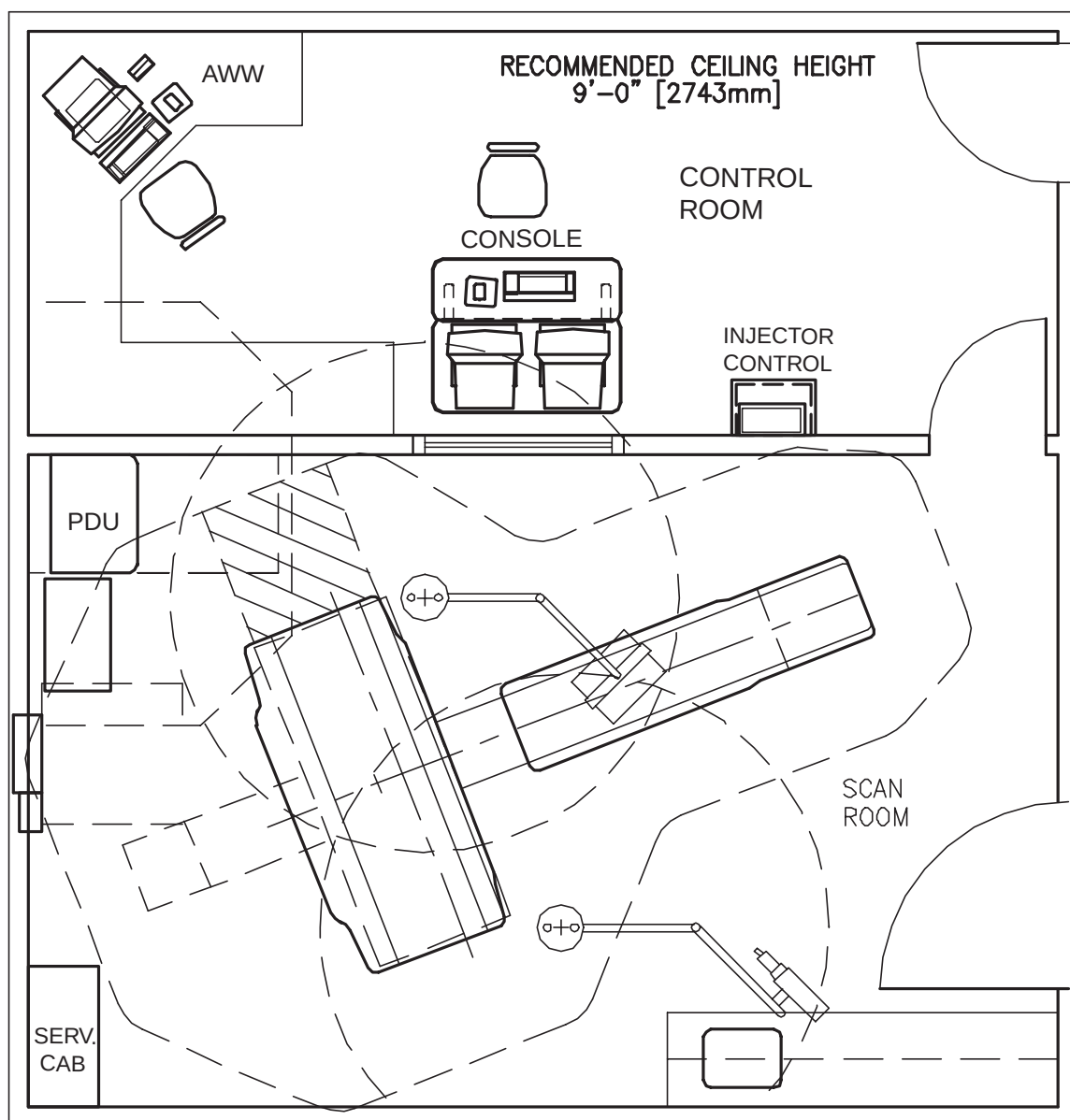


Figure 4-7 Typical Room Layout, with Injector Option

5.2 Storage Cabinet

A storage cabinet is provided by GEMS to store all supplied service equipment (see [Table 4-6](#) for equipment list). This storage cabinet (24" D x 26" W x 42" H) should be located in the scan room suite area, for easy service access. Refer to [Figure 4-6](#) or [Figure 4-7](#).

ITEM	SIZE	WEIGHT (TOTAL)
QA Phantom (water filled)	20 x 15 cm	12 lb
35CM Phantom	35 x 7 cm	18 lb
48CM Phantom	48 x 7 cm	25 lb
Phantom Holder	25 x 25 cm	8 lb
FE Box (Purple)	30 x 38 x 30 cm	15 lb
Rear Cover Dollies	158 x 82 cm	25 lb
Front Cover Dollies	85 x 20 cm and 85 x 15 cm	35 lb
Install Support Kit (box)	30 x 30 x 38 cm	20 lb
Tube Hoist Kit	77 x 8 cm and 38 x 15 cm	20 lb

Table 4-6 Equipment to be stored in storage cabinet

5.3 Advantage Windows Workstation (AWW)

Refer to Pre-Installation Manual 2111833 and Installation/Service Manual 2111831. Refer to [Figure 4-7](#) for sample layout of AWW option.

5.4 Control Room Considerations

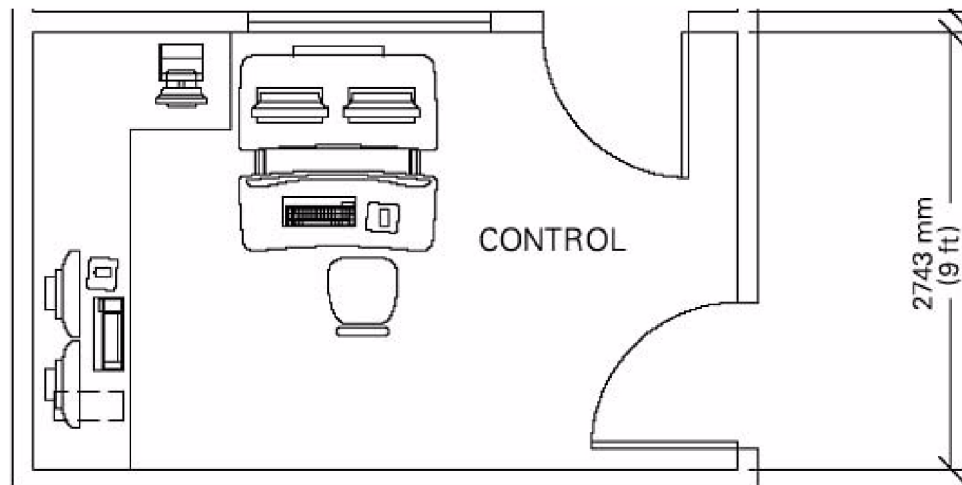


Figure 4-8 Typical Control Room Layout

- The control room must provide a suitable operating environment for the operator console electronic, and operator working comfort.
- The operator console cannot be dismantled or have components removed or rearranged in configurations other than as shipped.
- If operationally possible, the monitor desktop and user desktop components may be removed and placed on a counter-top, providing the cable lengths shipped are not altered or changed.

The operator console cabinet then can be remote mounted, provided the cooling requirements are met. Maintain 152 mm (6 in.) on all sides; and venting is required.

- A suitable work area, which is within reach of the operator console, should be provided for placement of the injector control. Injector controls differ in dimensions depending on the brand selected.
- A PACS, workstation, image printer, or filming device are often placed in the operator console control room area, and sometimes may be directly linked to the operator console.
- Additional components although linked via network or ethernet cable, are not powered from the CT operator console.
- Additional room power and network connection should be considered when reviewing the operator console work space.

Section 6.0

Component Dimensions

DESCRIPTION	WIDTH		DEPTH		HEIGHT	
	INCH	MM	INCH	MM	INCH	MM
System						
Gantry	86.6	2200	39.4	1000	74.6	1895
Table	24.3	617	94	2387		
Power Distribution Unit (NGPDU)	28	711	22	559	42	1067
Operator's console/computer	48	1219	39	991	33.5	851
Remote Color Monitor						
Color printer	23	584	18	457	7	178
Advantage Windows						
Workstation	16.5	419	16.1	409	3.1	79
Monitor	19.5	495	19	483	18.6	471

Table 4-7 Dimensions of Components

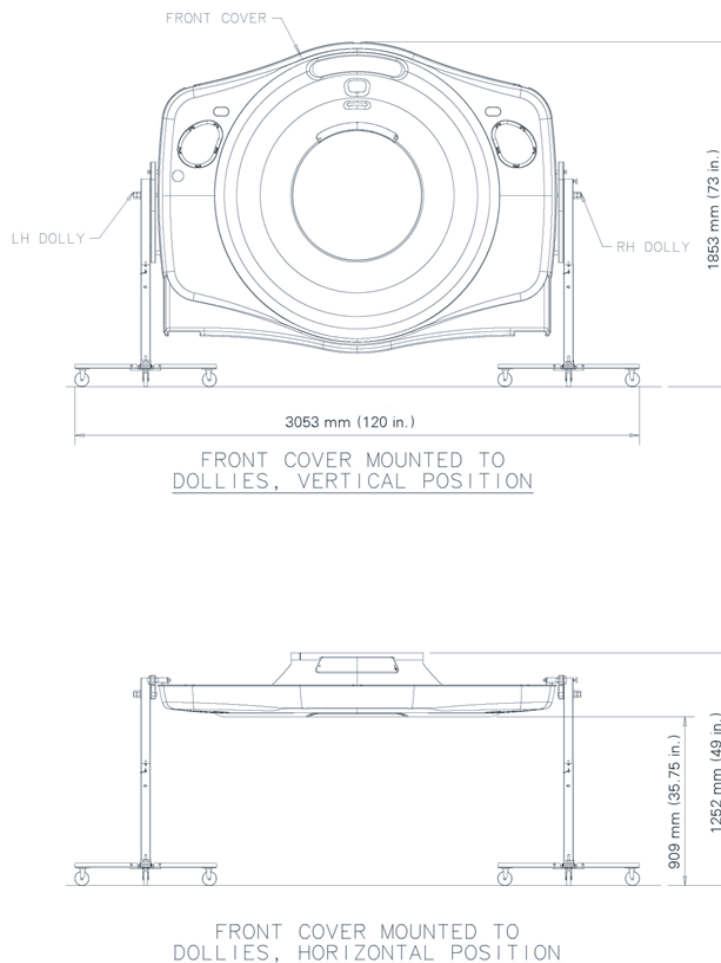


Figure 4-9 Gantry Front Cover with Service Dolly Dimensions

6.1 Table and Gantry

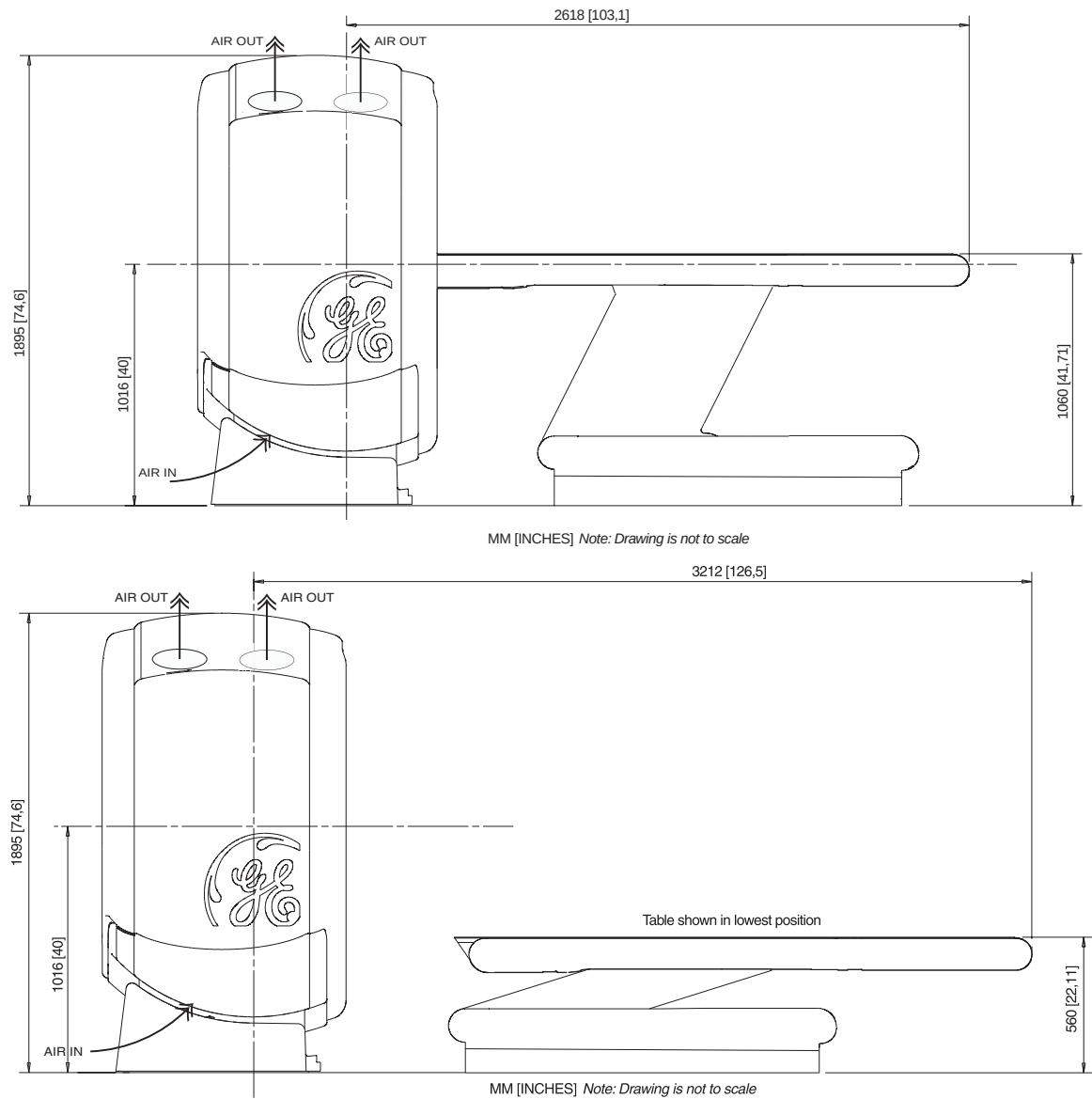


Figure 4-10 Table and Gantry (Side View)

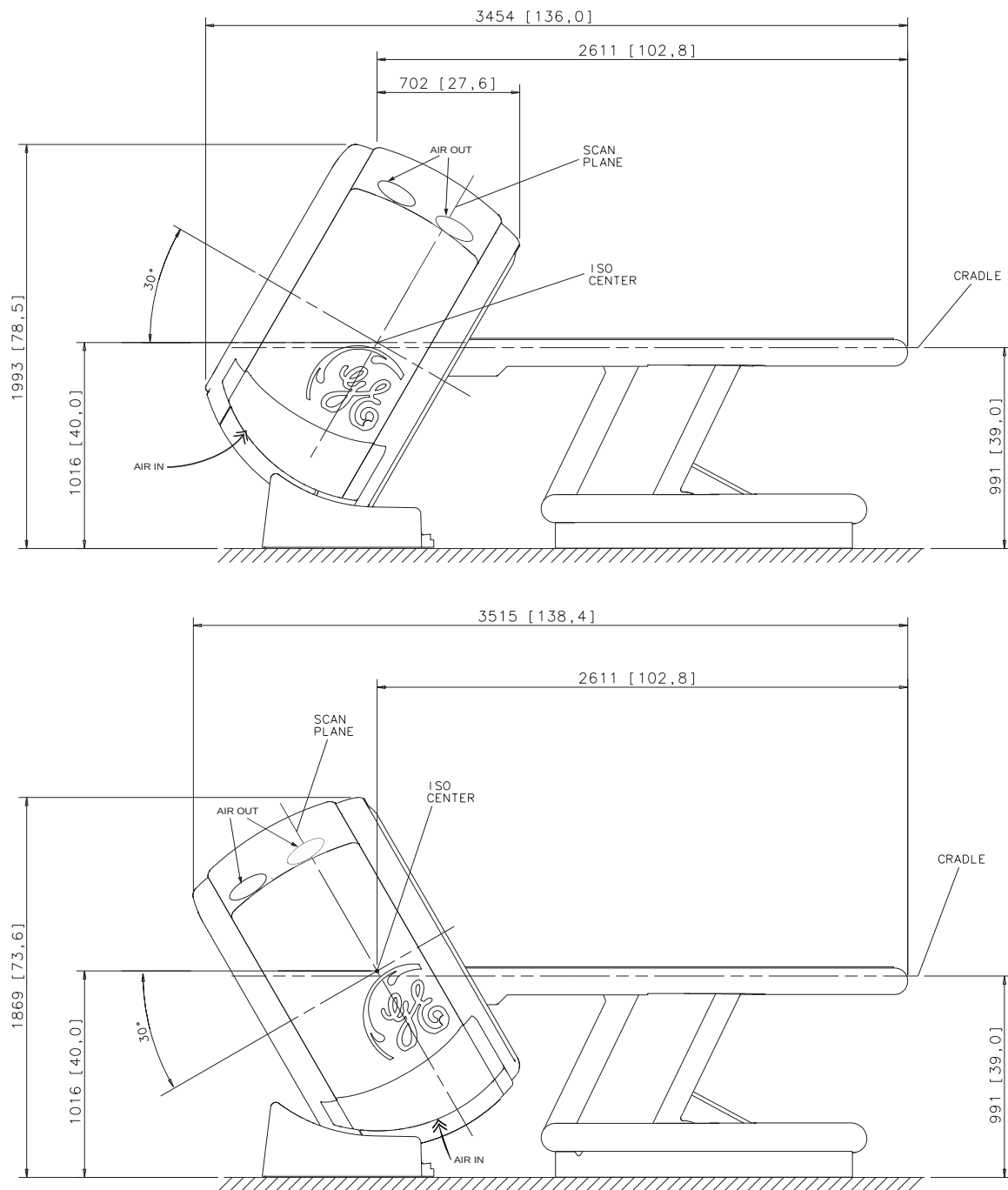


Figure 4-11 Gantry shown tilted +30° (top) and -30° (bottom)

6.2 Power Distribution Unit

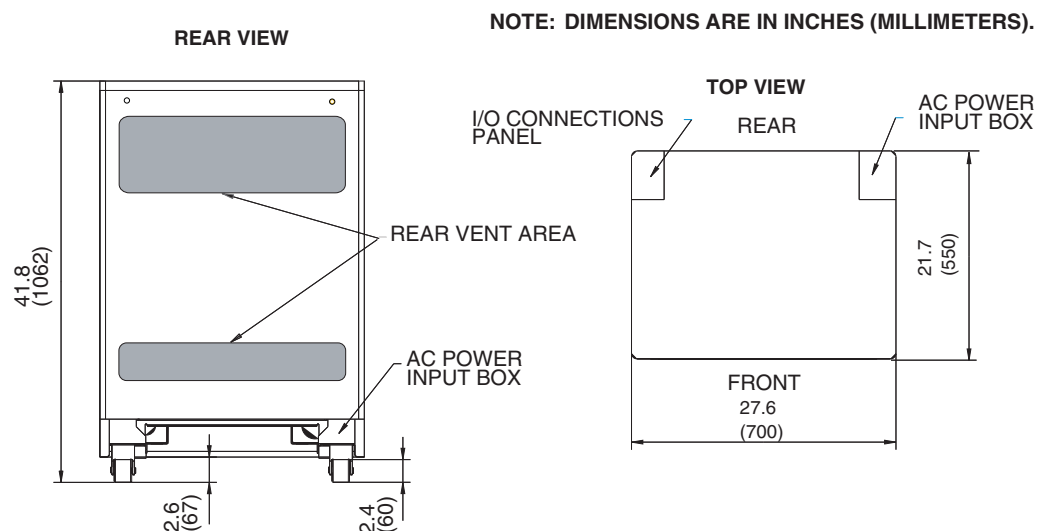


Figure 4-12 Power Distribution Unit (NGPDU)

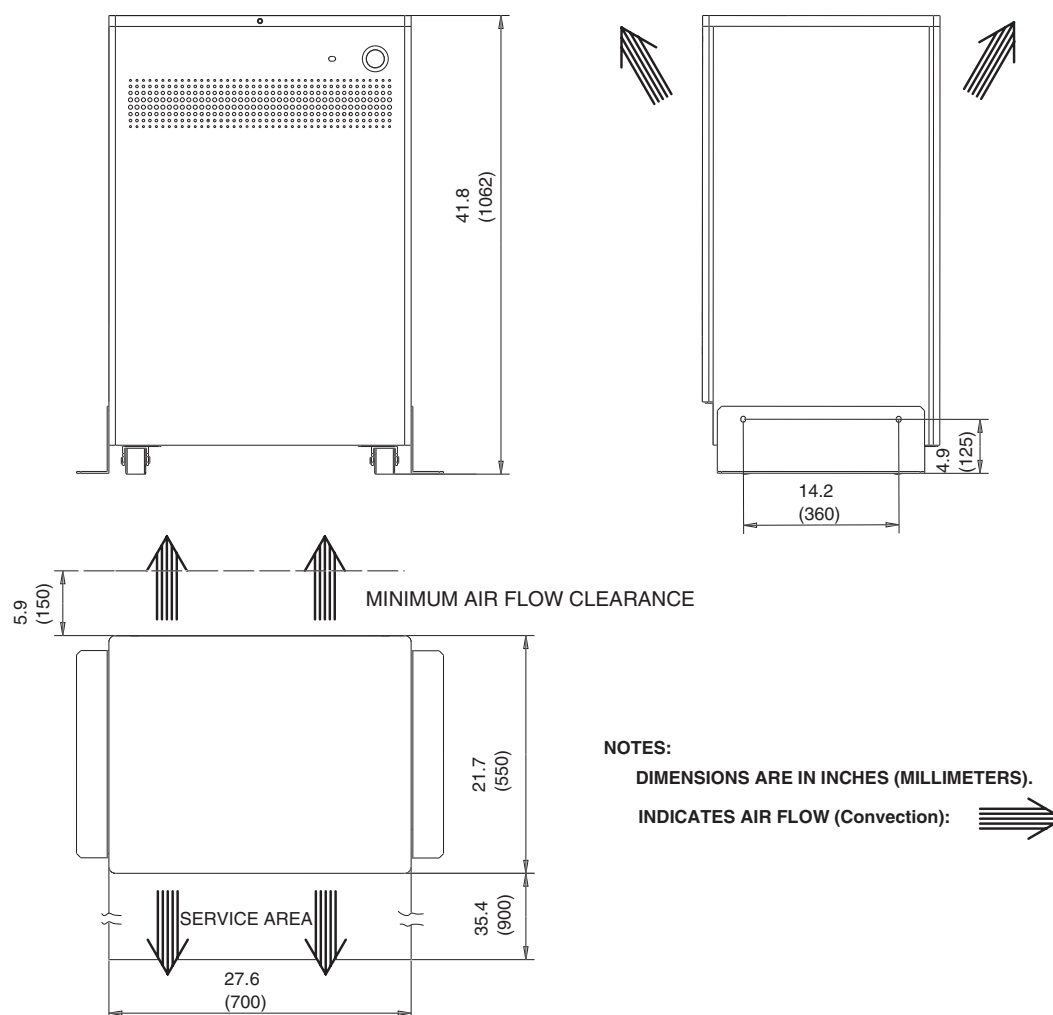


Figure 4-13 Power Distribution Unit (NGPDU)

6.3 Operator's Console

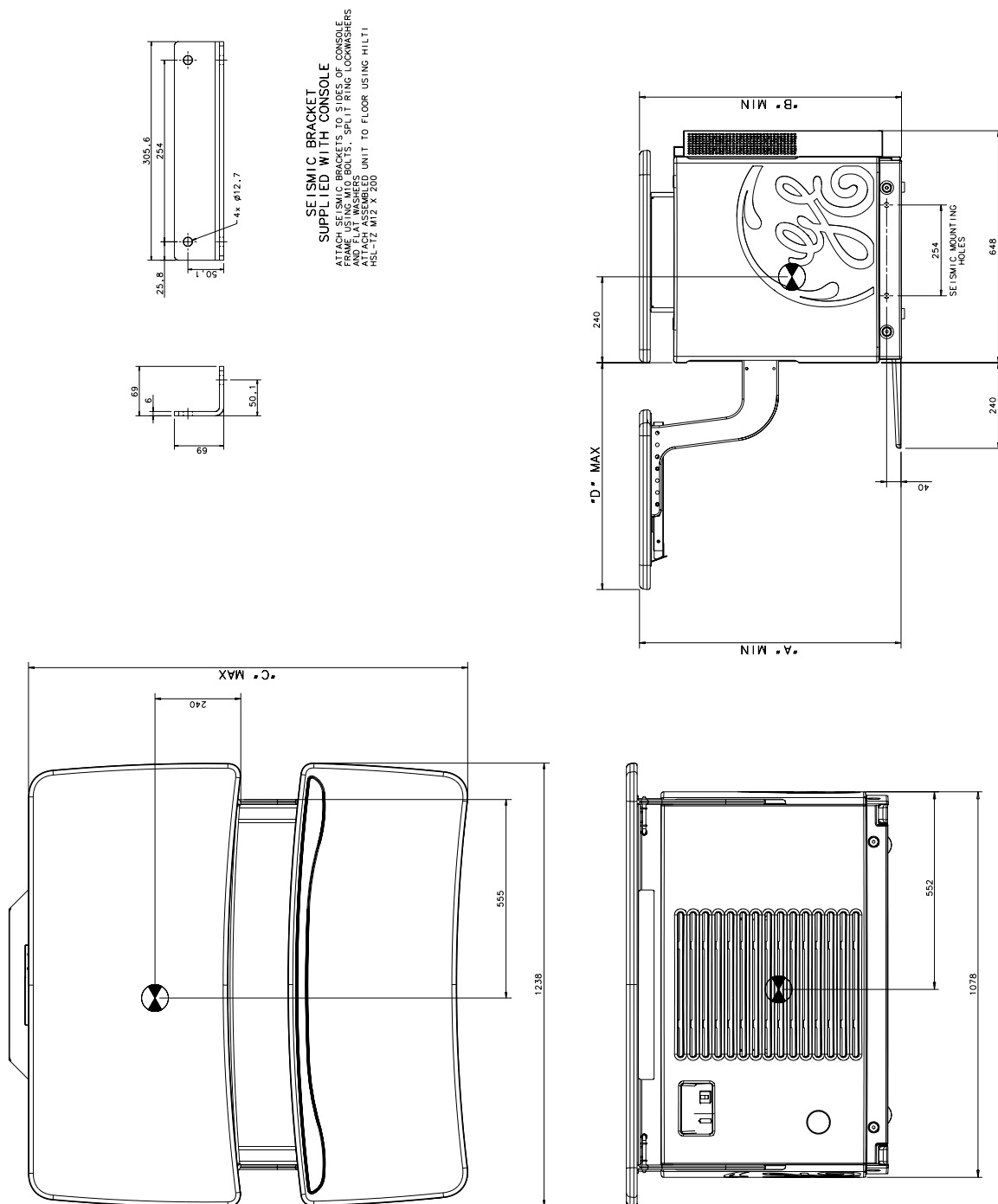


Figure 4-14 Operator's Console

Section 7.0 Structural Requirements

7.1 Suggested Ceiling Heights

A ceiling height of 108" (2743mm) is recommended. A ceiling height of 90" (2286mm) is the absolute minimum acceptable height for gantry installation.

7.2 Table and Gantry Mounting Requirements

Note: It is the purchaser's responsibility to provide an approved support structure and mounting method. General Electric is not responsible for any failure of the support structure or method of anchoring.



NOTICE LightSpeed 5.X scanners should be installed on 4" minimum thickness dense concrete floors. All other floor types must be approved by GEMS Engineering.

Table and gantry mounting dimensions are shown in [Figure 6-2](#), [Figure 6-3](#), [Figure 6-4](#). Refer to Chapter 6 for additional details of floor loadings, component weights, and Gantry and Table installation and anchoring.

Anchor gantry and table to floor by a means that will maintain their relative alignment and meet applicable building and other local codes, including seismic structural mounting requirements.

Floor structure must be capable of withstanding the occupied weight of table and gantry, and the individual contact area loading of these components. **The table-gantry requires a floor support in the 100 lbs/sq ft. (490 Kg/sq m) class or above. Any installation on a floor with a rating less than 100 lbs./sq. ft. (490 Kg/m²) should be braced to bring it up to this requirement.** Localized bracing to support the concentrated loads at the floor contact sections should also be provided.

Support areas of the patient table and gantry must rest on solid concrete, not resilient tile or carpeting which will slowly yield over a period of time and disturb alignment of table to gantry.

Factors that could cause misalignment between gantry and table due to floor sag should be considered. The cradle can potentially carry a 450 lb (205 kilogram) patient. Center of gravity changes as cradle cantilevers.

Take into consideration all other moving weights such as gurneys or personal equipment. Refer to Chapter 6 for gantry and table mounting details.

No part of floor surface within table and gantry, nor the two interface areas between table and gantry, should be higher than the support area for table and gantry.

A qualified person must verify that the site and method of anchoring are adequate to support loads and maintain table-to-gantry alignment. Location of supporting beams and columns may dictate position of table-to-gantry assembly. Use of flush floor duct or conduit in the floor may significantly affect floor strength.

The method and placement of anchoring through bolts must not reduce structural strength of floor.

7.3 Floor Strength

Concrete floors must have a minimum strength of $f'_c = 2000 \text{ psi}$ ($1.4 \times 10^7 \text{ Pa}$) at 28 days for mounting floor anchors. It is the responsibility of each customer to have appropriate tests performed to determine and measure concrete strength. Your GE Service representative can assist you.

Consult GE Installation Support Services for further details.

7.4 Floor Levelness

The CT Room floor levelness requirement is important for accurate patient positioning. Floor levelness in the Scan Room must not be greater than 0.3125 (8mm) between depression and high spots over any 120 in. (3048 mm) distance within the area of the gantry and the area around the table (see the envelope shown in [Figure 4-6](#) or [Figure 4-7](#)).

The use of floor shims is not suitable to achieve floor levelness. It is recommended that the concrete be leveled to meet this requirement.

7.5 Floor Vibration

The CT equipment may be sensitive to vibration in the frequency range of 0.5 to 20 Hz depending on the amplitude of the vibration. It is the customers responsibility to contract a vibration consultant or qualified engineer to implement design modifications to meet the specific limits, However, it is ultimately the customer/architect/engineer responsibility to design the site solution.

7.5.1 Steady State Vibration

The maximum steady state vibration transmitted through the floor should not exceed 10^{-3} m/s² rms maximum single frequency above ambient baseline from 0.5 to 80 Hz (measured in any 1 hour during a normal operating period).

7.5.2 Transient Vibration

The behavioral characteristics must be such that any measurable transient disturbance must also be minimized to less than 0.01 m/s² peak-to-peak.

7.5.3 Equipment Location

To minimize the interference, the CT equipment should be placed on a solid floor, located as far as possible from the following vibration sources:

- Parking lots
- Roadways
- Subways
- Trains
- Hallways
- Elevators
- Heliports
- Hospital power plants containing pumps, motors, air handling equipment and air conditioning units

7.6 Walls: Scan Window

The recommended patient viewing window dimensions are 48 in. wide by 42 in. high (1219 mm x 1067 mm). The location of the window is dependent on the position of Operator Workspace position (see [Figure 4-6](#) and [Figure 4-7](#)).

Note: The operator at the Operator Workspace must be able to view the patient during a scan.

7.7 Mobile Units

It is ultimately the responsibility of the customer/architect/engineer to design the site solution (i.e., levelness, vibration, equipment location and proximity effects of surrounding equipment, such as

other mobile, transportable and relocatable units that could contain CT, MR, or other medical devices).

Section 8.0

Network Connections

Broad-band is considered the standard network connection for LightSpeed 5.X. (A dial-up modem is optional.) Broad-band connections should use one of the following Category 5 patch cables:

CAT Num	GE Part Num	Length
K9000WB	2215028-10	20 m
K9000KP	2215028-5	10 m
K9000JR	2215028-4	5 m
K9000WA	2215028-9	3 m

- The CT system is connected to the network through the Console.
- A patch cable (not to exceed 10 feet) should be provided by the customer, and it is used to connect the console to a wall box. (See Notes on [Figure 9-3](#))
 - Some customer-site units may require cable duct-work or conduit to route connecting network cables to the workstation, camera and console.
 - The run from the hospital switch to the CT wall outlet must not exceed 290 ft. (88m). Bandwidth performance is degraded when the length reaches 300 ft. (91m) or greater.
 - For the optional modem: **Two phone lines should be provided by the facility.** One line is for use with a modem and must be an analog line. The second line is a voice only line.

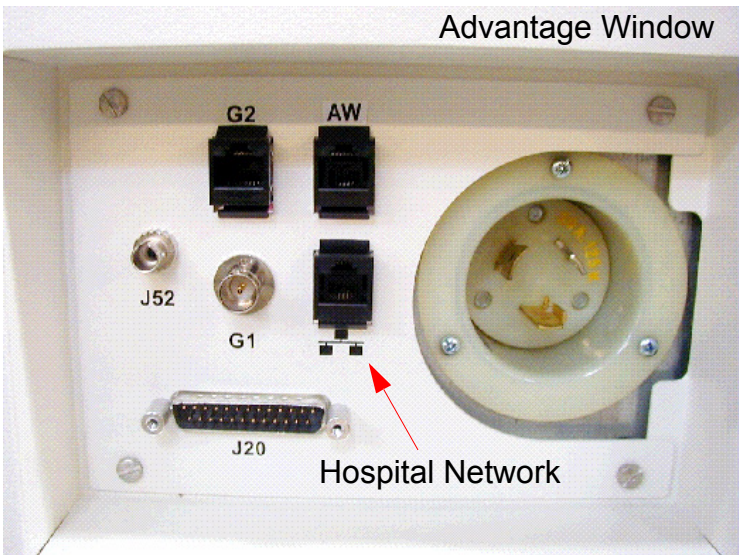


Figure 4-15 Console Rear Bulkhead

8.1 US Process Overview

The United States network connectivity requirement for this product is broad-band. The US process relies on the Install Specialist to select a Customer Champion and identify an IT contact for the site. Together, those individuals then complete a site assessment to gauge what tasks are needed to fulfill the connection.

Anyone can contact the GE Connectivity team at 800.321.7937, Option #3, with questions.

8.1.1 Customer Broad-Band Responsibilities

Provide GEMS Installation Specialist with an accurate site address, telephone number, contact name, and email address for the:

- Customer Champion
 - Co-ordinate VPN activities between Radiology/Cardiology and the Information Technology (IT) departments
 - Act as a focal point in assuring site broad-band infrastructure meets GEMS requirements for connection as determined by a mutual assessment with the GEMS Connectivity team.
- IT Contact
 - Complete an equipment assessment with GEMS Connectivity team to determine site readiness for broad-band
 - Work with the Customer Champion to complete any identified infrastructure changes
 - Provide IP addresses for new CT equipment
 - Provide a VPN compatible appliance that will support the IPSec tunneling protocol and 3DES data encryption
 - To utilize an Internet Service Provider that supports static routing



GE Medical Systems

Remote Service Broadband - Customer Site Assessment

Site Name: _____ FE Name: _____
City, State: _____ FE Phone: _____
Date: _____ FE Email: _____

- | | Yes | No |
|---|---|--------------------------|
| 1. Does your site currently have a persistent (24x7) Internet connection? | ☐ | ☐ |
| 2. Is the GEMS Diagnostic Imaging equipment on the Local Area Network and will it be accessible to the Internet? | ☐ | ☐ |
| 3. Does your site have a VPN device today? | ☐ | ☐ |
| 4. Is the VPN device one of the models below? <i>If Yes, please select the model from the options below.</i> | ☐ | ☐ |
| ☐ a) Cisco Pix Firewalls
☐ b) Cisco Routers
☐ c) Cisco 3000 Series (Altiga acquisition)
☐ d) Checkpoint Firewalls Software Version 4.1 and higher
☐ e) Nortel Contivity Software Version 3.2 or higher
☐ f) Redcreek | ☐ g) Symantec (Raptor) firewalls
☐ h) Firebox
☐ i) Linux S/WAN
☐ j) Sidewinder
☐ k) Netscreen
☐ l) None
☐ m) Other _____ | |

**If No, the GEMS Connectivity Support Team can help determine device compatibility.*

5. Does your VPN device support "triple DES" Encryption? ☐ Yes ☐ No
6. Has approval been given to install this VPN connection? ☐ Yes ☐ No
- Site Approver's Name _____
7. Provide your VPN Installer information, this is the person who will be contacted to schedule the VPN install.

Customer Installer Name: _____
Installer Telephone Number: _____
Installer e-mail address: _____

Notes: _____

Field Engineer needs to provide compatible system information:

System ID	IP Address	Gateway Address
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____

[Additional System and IP Address Spaces Available on Page 2](#)

If you have questions or need assessment support contact your Zone Champ or:

Joe Gracz - HQ Support 1-262-524-5261
Joseph.Gracz@med.ge.com

Once you have completed both pages of this form, please send it to:

a) Judy Heyer judy.hey@med.ge.com
b) Judy Heyer Fax# 414-918-4707

[Use the send button on page 2](#)



GE Medical Systems

[illegible]

Once you have completed this form, please send it to:

Judy Heyer
Judy Heyer

email
Fax #

judy.heyer@med.ge.com
414-918-4707

☐ Send Now

Section 9.0

Radiation Protection

Figure 4-16 depicts measurable radiation levels within the scanning room while scanning a 32 cm CTDI phantom (body) and a 20 cm water phantom (head) with the technique shown.

Note:
Actual
measurements
can vary.

All measurements have an accuracy of ± 20% because of measurement equipment, technique, and system-to-system variation.
Use the correction factors shown in Figure 4-8 to adjust exposure levels to the usual scan technique at your site.

CHANGED PARAMETER	MULTIPLICATION FACTOR
mAs	new mAs/100
80 kV	0.21
120 kV	0.71
140 kV	1.0
4 x 3.75mm images	0.82
<u>16 x 0.625 LD</u> <u>8 x 1.25 LD</u> <u>4 X 2.5 LD</u> <u>Fluro 5mm</u>	<u>0.59</u>
4 x 1.25 LD 5mm (1i) Fluro 2.5 mm	0.40
1 x 1.25mm images	0.20
2 x 0.625 LD 1 x 1.25	0.10

Table 4-8 Shielding Requirements Scaling

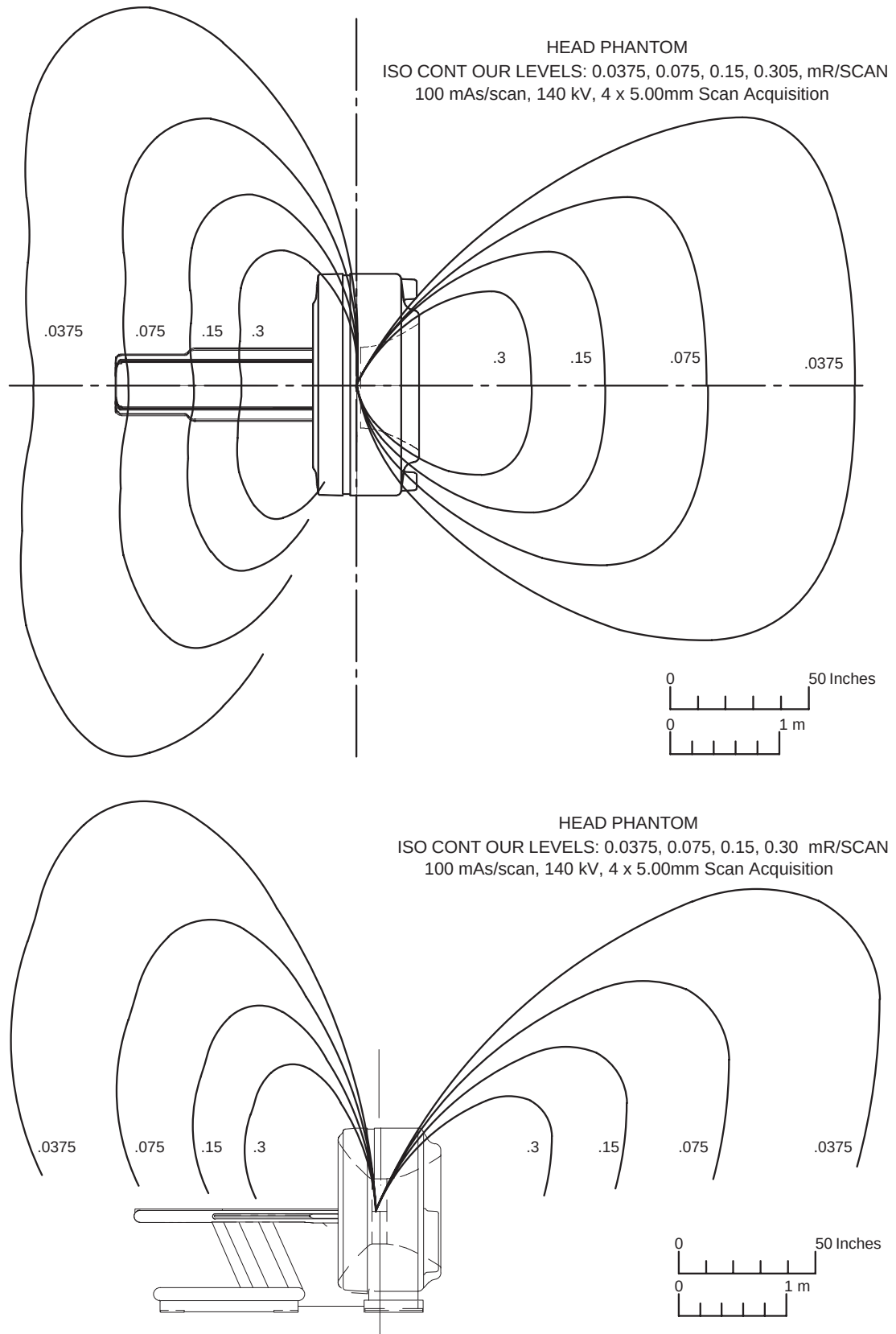


Figure 4-16 Typical Scatter Survey (Head Filter)

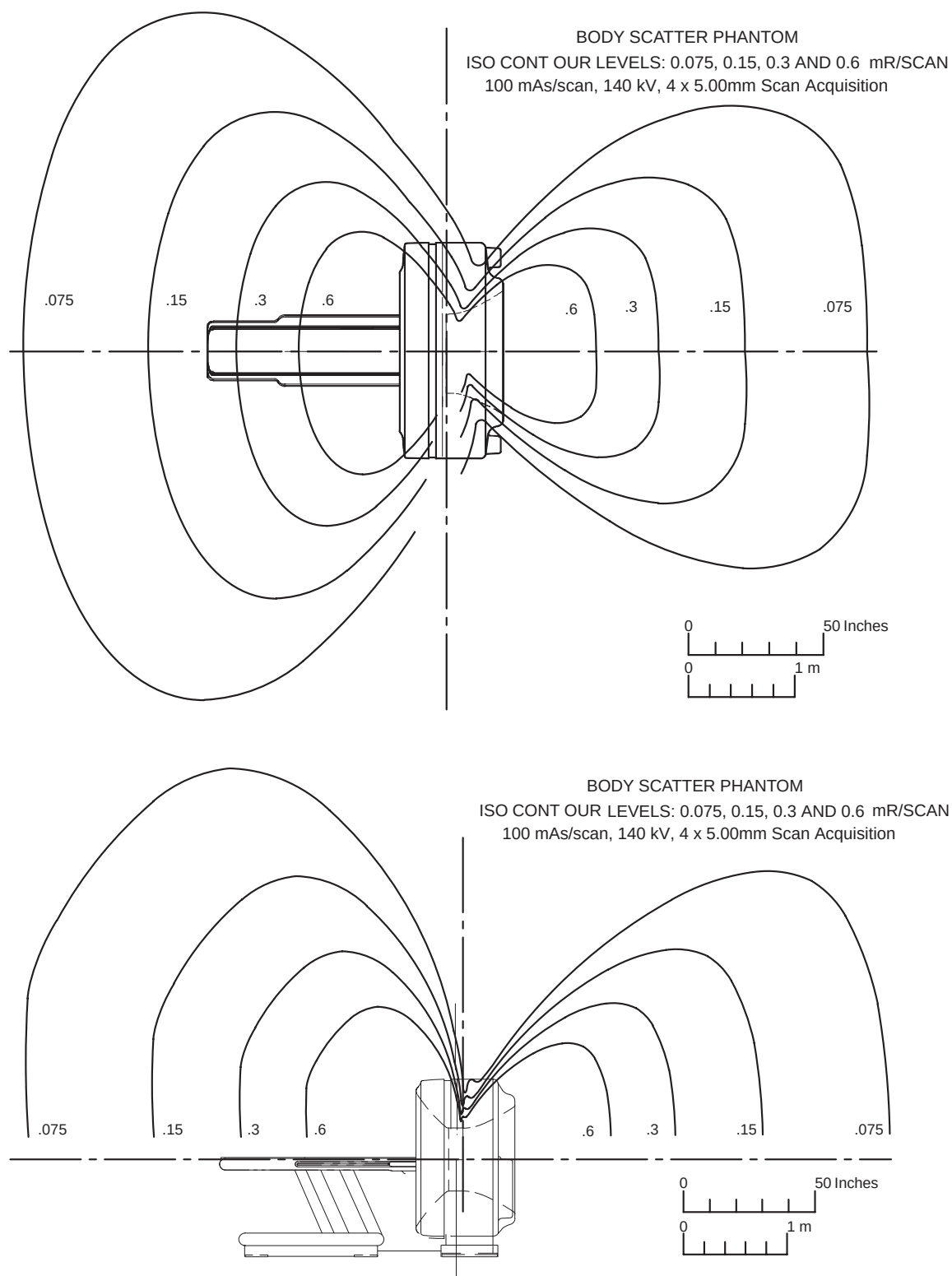


Figure 4-17 Typical Scatter Survey (Body Filter)

Chapter 5

Environmental Conditions

Ratings and duty cycles of CT subsystems apply if site environment meets the standards of this section. Maintain environmental conditions listed below at all times – including, for example, overnight, weekends and holidays. Shut down the CT system if air conditioning is not working. When system is shut down for major repair, air conditioning may be shut down also.

Section 1.0

Temperature and Humidity Specifications



NOTICE
Potential
Equipment
Failure

Do not operate (i.e., “Power ON”) Gantry or Console subsystems with ambient room temperatures exceeding 75° F. Scan room or control room temperatures in excess of 75° F can result in the failure of gantry or console components.

- Ambient Temperature: (Fahrenheit and Celsius)
 - Scan Room: Maintain a temperature of 72° F $\pm$ 3° F (22° C), for patient comfort. When scan room is unoccupied, table and gantry temperature limitations are 70° - 75° F (21° - 24° C).
 - Control Room (including Console/Computer): Maintain 72° F (22° C).
 - Equipment Room: If a separate equipment room is used to house the PDU, the allowable temperature range is 60° - 75° F (15° - 24° C).
- Store media (cartridges) in long-term storage in same temperature range as host computer.
- Store media in the host computer environment for one-half hour before use.
- Maintain relative humidity of 30%-60% (non-condensing) during operation (all areas).
- The maximum temperature rate of change is 5° F/hr. (3° C/hr.).
- The maximum relative humidity rate of change is 5% RH/hr.

Section 2.0

Temperature and Humidity Monitoring

Locate computing subsystems in an area that has the environment specified in [Section 1.0](#), above. First, assess your environment's heat and humidity. If necessary, temporarily install a temperature and humidity recorder close to where the gantry will be installed. Note readings before installation and again after installation to verify true temperature and humidity conditions for your environment. Consider your HVAC needs and redundancy. You may wish to consider an air conditioner with two compressor units rather than one. A backup (redundant) air conditioner permits CT system operation during an extended repair of the primary air conditioner.

Section 3.0

Cooling Requirement

Use [Table 5-1](#) to assist in cooling requirements planning. Over half the cooling used by your scanner is required for gantry operation. Locate a wall air-conditioning vent at floor level beside and behind gantry to meet both gantry cooling needs and provide patient comfort. Do not locate any cooling vents directly above the gantry. Air returns above the gantry are recommended.

SYSTEM COMPONENT	BTU/HR	WATT
1. Gantry recommended (See NOTE 1)	25,100	7,350
2. Table	700	200
3. Power distribution unit	3400	1000
Recommended Scan Room Subtotal (see notes):	29,200	8,550
4. Operator's console/computer with one IG	7361	2165
Additional IG, each	1360	400
Monitor, each (CRT)	2346	690
SCSI Tower	425	125
Injector	425	125
DASM	425	125
Recommended Control Room Subtotal (includes OC w/1 IG, 2 monitors & SCSI Tower):	12,342	3630
System Total (Recommended) (See NOTE 1)	41,542	12,180
Option: Remote Color Monitor	1178	345
Option: Advantage Windows	256	75
ROOM TOTAL (SEE NOTE 2)		
NOTE 1: With 75 scan rotations per patient: Recommended BTU/hr. provides for up to six patients per hour. It is also needed during calibration of the system.		
NOTE 2: Cooling requirements do not include cooling for room lighting, personnel or non-CT equipment.		

Table 5-1 Cooling Requirements (Worksheet)

Please refer to [Figure 4-10](#), [Figure 4-13](#) and [Figure 4-14](#) in [Chapter 4, Section 6.0](#), for component air flow requirements.

Section 4.0

Altitude

System operating altitude is from mean sea level to 10,000 ft. (3050 meters).

Section 5.0

Electro-Magnetic Interference (EMI)

5.1 Gantry and Table

Locate gantry in ambient static magnetic fields of less than 10^{-4} tesla (1,000 milligauss) to guarantee specified imaging performance. Ambient AC magnetic fields must be below 10^{-6} tesla (10 milligauss) peak.



WARNING REQUISITES FOR ROOM

GANTRY & TABLE ARE ALLOWED TO BE INSTALLED ONLY IN X-RAY PROTECTED ROOMS, WHICH PROVIDE AN ATTENUATION OF AT LEAST 12dB FOR RADIO DISTURBANCES FROM 30 MHz TO 1 GHz.

(The Warning statement above is specified by IEC 60601-1-2, First Edition, 1993)

5.2 Color Monitor

Locate color monitors in ambient static magnetic fields of less than 5×10^{-5} tesla (100 milligauss) to guarantee color purity and display geometry. See [Figure 5-1](#).

5.3 Console / Computer Equipment

Locate computer equipment in ambient static magnetic fields of less than 10^{-3} tesla (10,000 milligauss) to guarantee data integrity. See [Figure 5-1](#).

5.4 Magnetic Media

Locate magnetic media in ambient static magnetic fields of less than 10^{-3} tesla (10,000 milligauss).

5.5 PDU

The PDU produces an electromagnetic field that radiates outward from its cabinet in all directions. Do not place sensitive electronics (e.g., console or computer equipment - the UPS is not classified as sensitive electronics) within one-and-a-half meters (1.5m) of the Power Distribution Unit., in any direction (including above or below). See [Figure 5-1](#).

5.6 EMI Reduction

If fields of excessive EMI are known or suspected to be present, consult GE Medical Systems Sales & Service for recommendations. Consider the following if you attempt to reduce EMI:

- External field strength decreases rapidly with distance from source of magnetic field.
- External leakage magnetic field of a three-phase transformer is much less than that of a bank of three single phase transformers of equivalent power rating.
- Large electric motors are a source of substantial EMI.
- High-powered radio signals are a source of EMI.

Maintain good screening of cables and cabinets.

5.7 UPS

The Uninterruptable Power Supply (UPS) provides a consistent power supply to various electrical components of the system. Also, it continues to provide electrical power to components during a site-wide power outage so components can be safely shut down. The UPS should be kept at least one meter (1m) away from sensitive electronics (the PDU does not include sensitive electronics).

5.8 Equipment EMI “Envelopes”

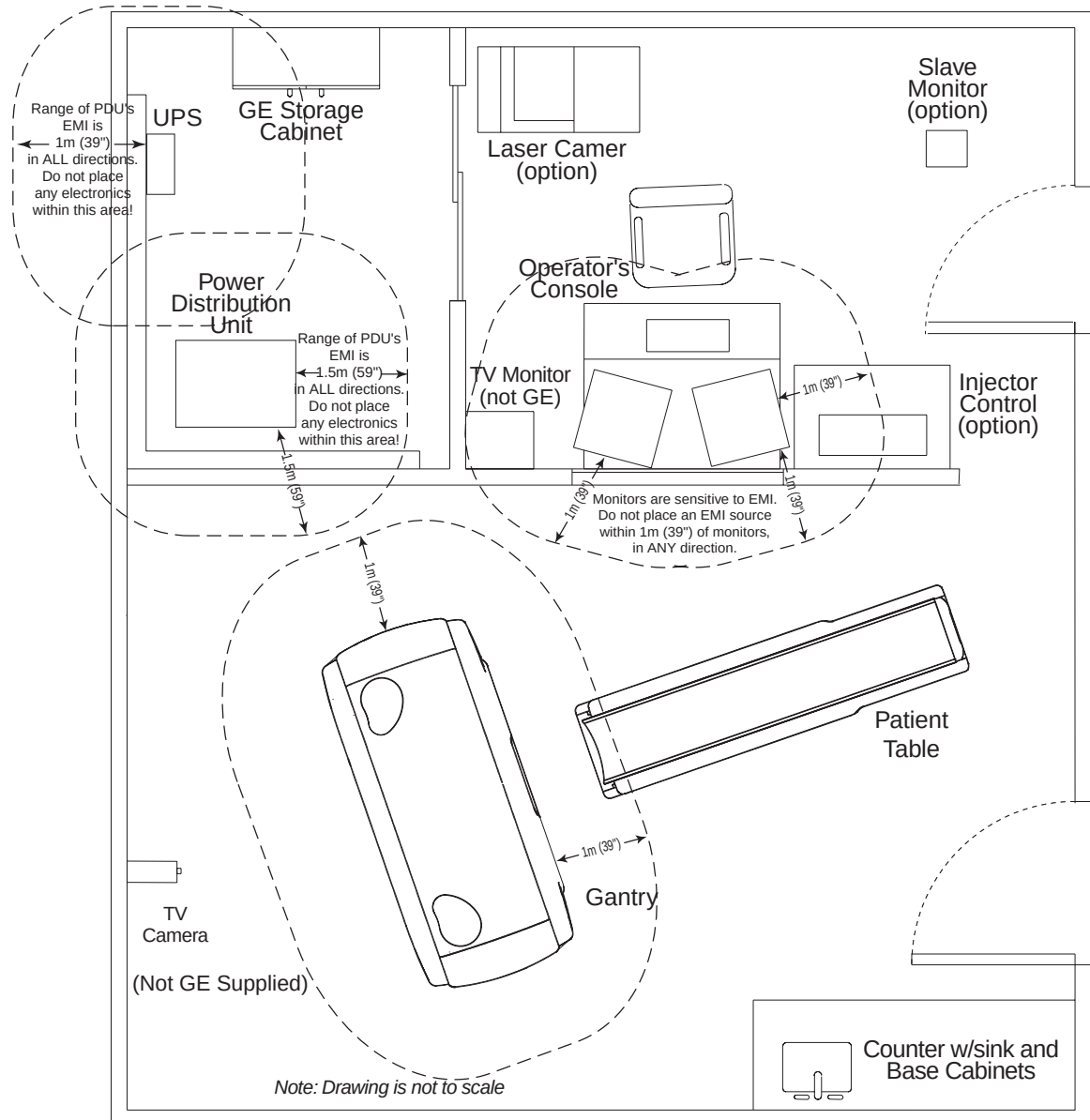


Figure 5-1 Sample Room Layout, showing approximate EMI requirements

Chapter 6

Floor Loading and Weights

Section 1.0 Floor Loading

The LightSpeed 5.X system has a total floor load of approximately 6500 lbs (2950 kg). About 175 lbs (2350 kg), including patient (450 lbs (204 kg)), is concentrated in the table-gantry assembly.

Table 6-1 lists CT components with weight, size, floor loading and normal mounting requirements.

ITEM	NET WEIGHT LB(KG)	OVERALL W X D INCH (MM)	WEIGHT/AREA LB/SQ. FT. (KG/M ²)	LOAD PATTERN IN. (MM)	NORMAL METHOD OF MOUNTING IN. (MM) (GE SUPPLIED) ¹
Gantry (no covers)	3770 (~1710)	86.6 X 39.4 (2200 X 1000)	268 (1309)	Rectangular base plate 27.5 X 79.3 (700 X 2012) with four round pads, each 2.5 (63.5) in contact with floor. Individual pad loadings are: 910 lb, 960 lb, 1040 lb, and 1090 lb (see Figure 6-3).	Hilti Kwik-Bolt II 1/2in (12.7mm) diameter by 8in (203mm) long per P/N 2106573 at four leveling pads into concrete floor.
Dollies (each)	250 (114)				
Top Cover (each)	24.6 (11.2)				
Side Cover (each)	25 (11.3)				
Front Cover	95 (43)				
Rear Cover	100 (45)				
Patient Table	1185 (538) Includes 450 (204) Patient	24.3 X 94 (617 X 2387)	175 (852)	Rectangular base 17 X 57.5 (432 X 1460) with five round pads, each 2.5 (63.5) in contact with floor. Worst-case cantilever pull on any bolt is 1235 lb (561 kg).	Hilti Kwik-Bolt II 1/2in (12.7mm) diameter by 8in (203mm) long per P/N 2106573 at three leveling pads into concrete floor.
Dolly	215 (98)				
Power Distribution Unit	~770 (~350)	28 X 22 (711 X 559)	185 (909)	Four Casters support area of 28 X 22 (711 X 559).	Casters are for positioning and service. Set on floor. May be anchored to floor using angle brackets ² in seismic zones.
Console w/HP & w/o monitors	450 (204)	48 X 39 (1219x991)	140 (681)	Four Casters or Leveling Feet support area of 46 X 19 (1168 X 483).	Casters are for positioning. Set on floor. Console may be anchored to floor using angle brackets ²
Monitor - CRT (ea)	80 (36)				
Monitor - LCD (ea)	22 (10)				

Notes:

- 1.) Use the GE Supplied mounting hardware ONLY IF APPROVED by qualified personnel.
[See statements in 7.2 - Table and Gantry Mounting Requirements, on page 63.]
- 2.) Angle brackets are included on the shipping skid, and are also available in the OPTIONAL Seismic Kit (R4390JC).

Table 6-1 LightSpeed 5.0 System Floor Loading

Section 2.0

Mounting Data, Including Seismic

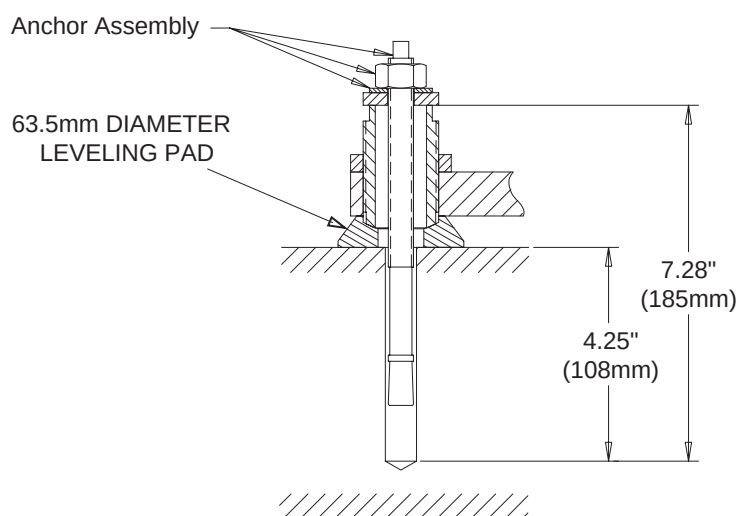
The following pages show center-of-gravity information for system components:

- Gantry: [Figure 6-2](#)
- Table: [Figure 6-4](#)
- Power Distribution Unit: [Figure 6-5](#)
- Operator's Console/Computer: [Figure 6-6](#)

Floor mounting hole locations for components that don't have templates are also in this section.

An optional Seismic Kit (R4390JC) is available, for facilities in areas that require one.

Customer is responsible for seismic mounting. Refer to all applicable codes in your area.



Note: Adjusters are used at each anchor location. Anchor hole ID is 1" (2.5 cm).
Void between adjuster and anchor must be filled according to local building codes for seismic application.

Figure 6-1 Typical Floor Anchor, Gantry and Table

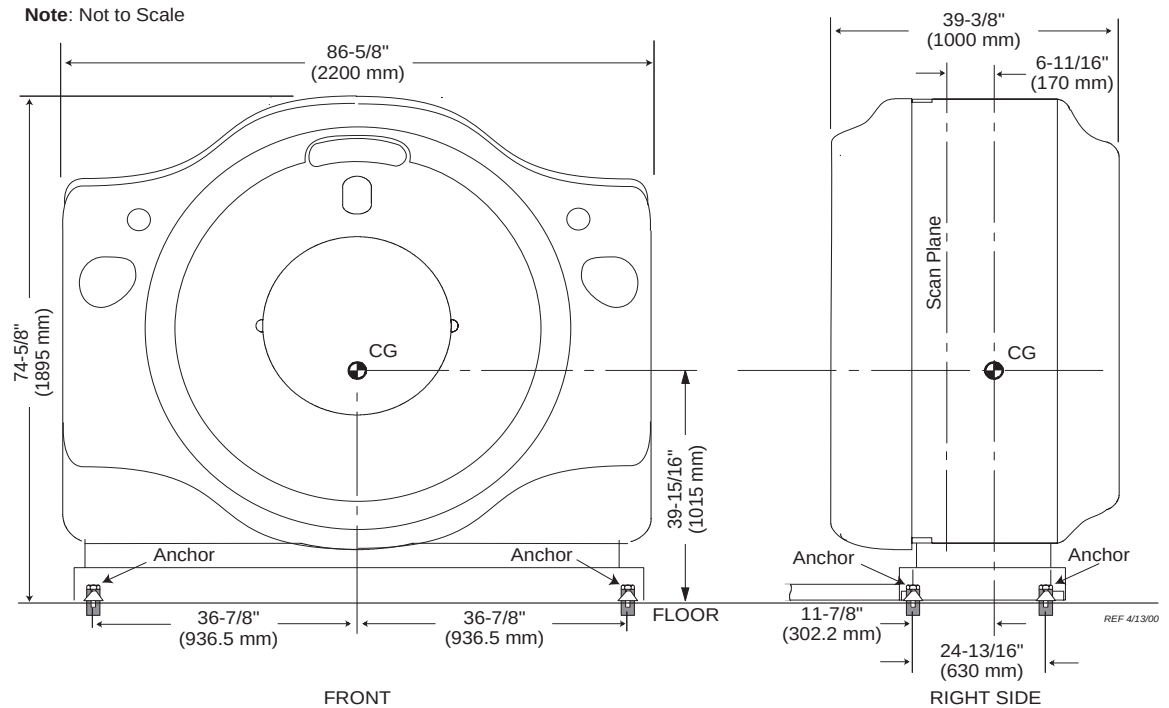
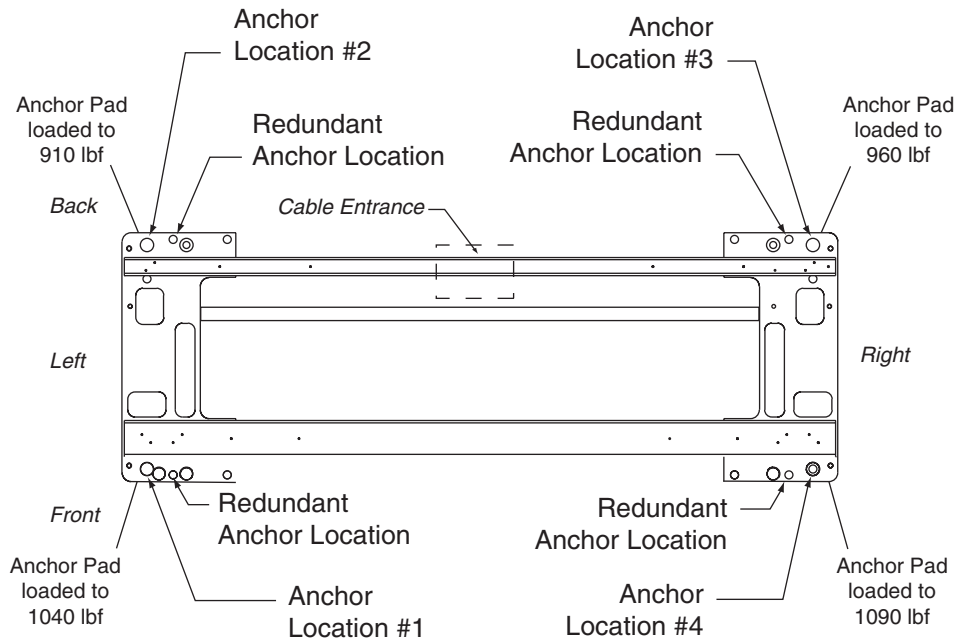


Figure 6-2 Gantry (CT2)



Note: Adjusters are used at each anchor location. Anchor hole ID is 1" (2.5 cm). Void between adjuster and anchor must be filled according to local building codes for seismic application.

Figure 6-3 Gantry Anchor Locations

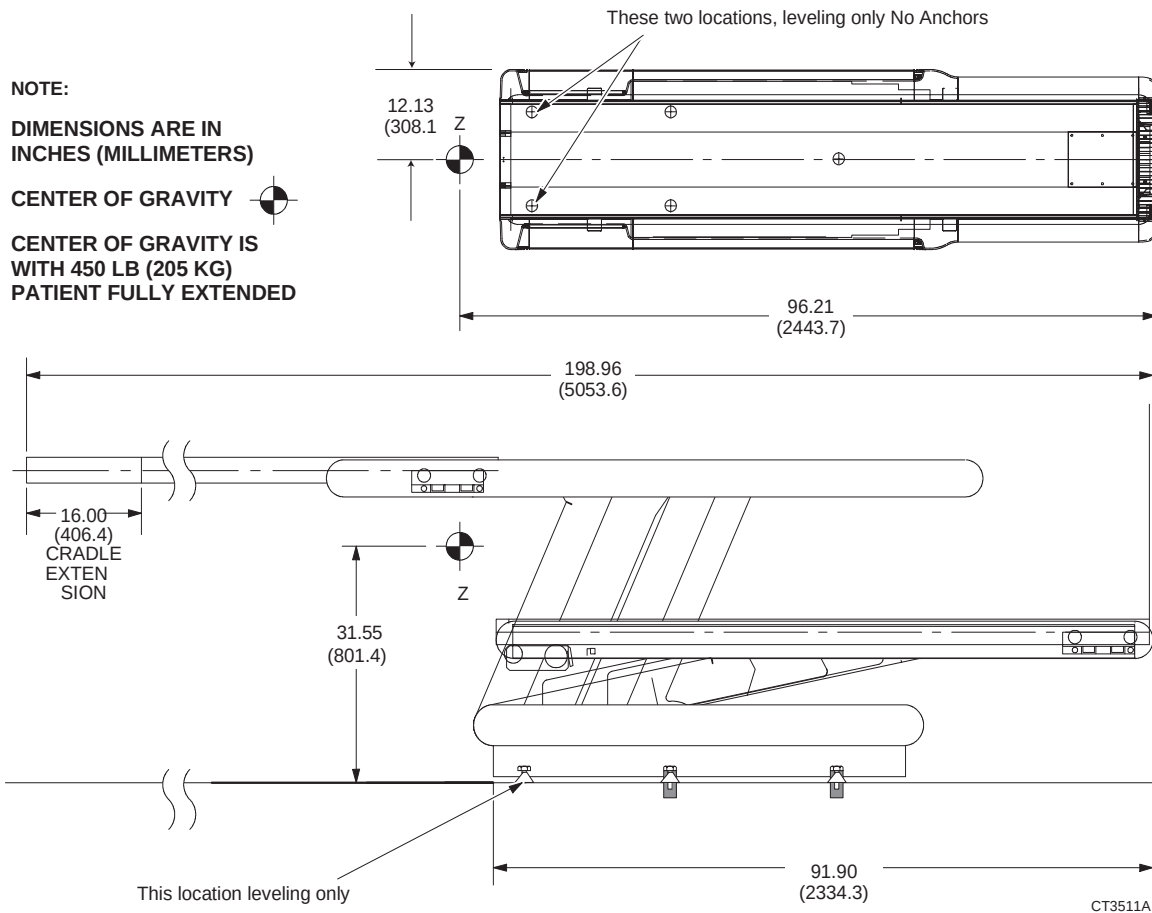


Figure 6-4 Patient Table (CT1)

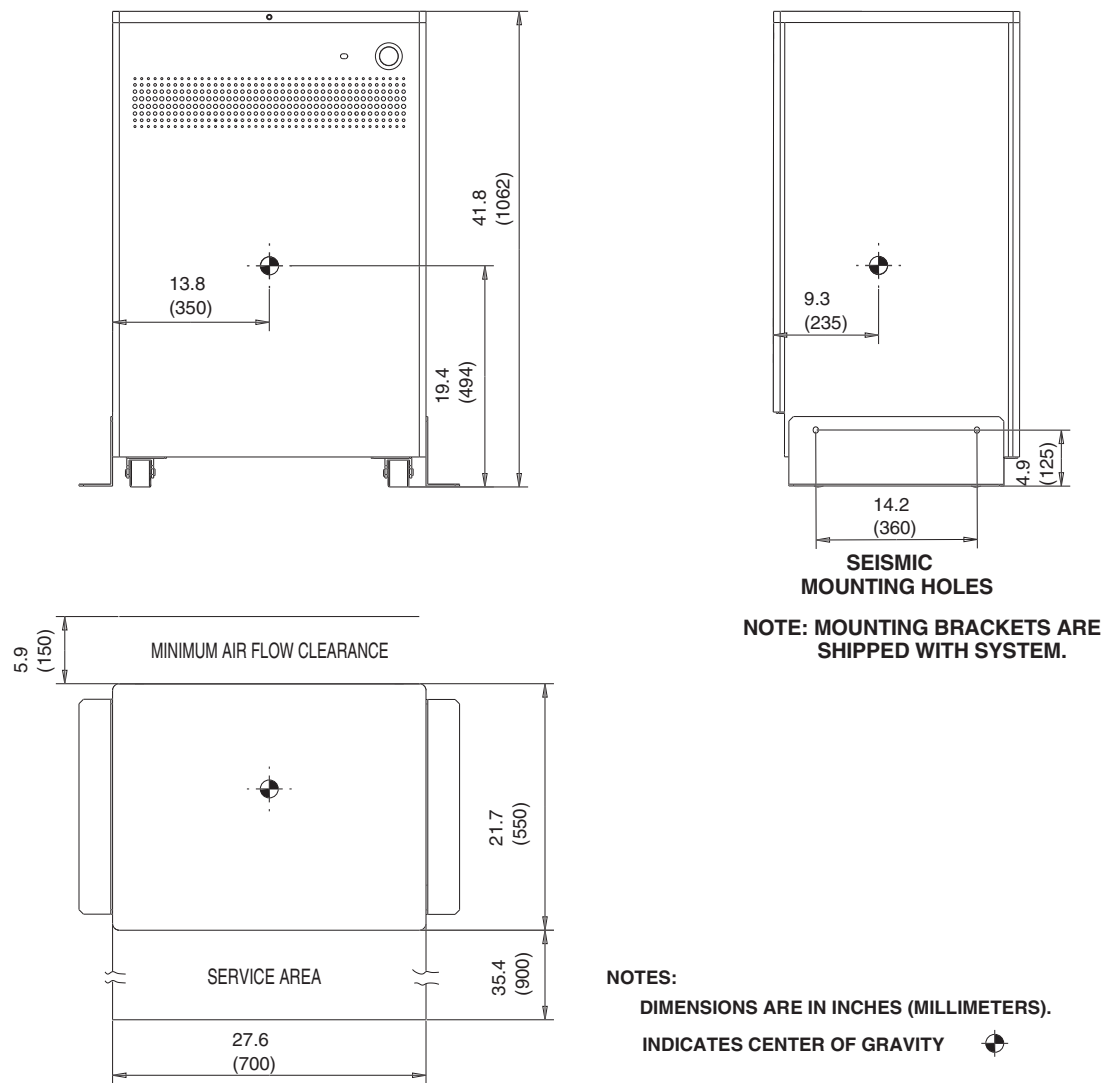


Figure 6-5 Power Distribution Unit (NGPDU)

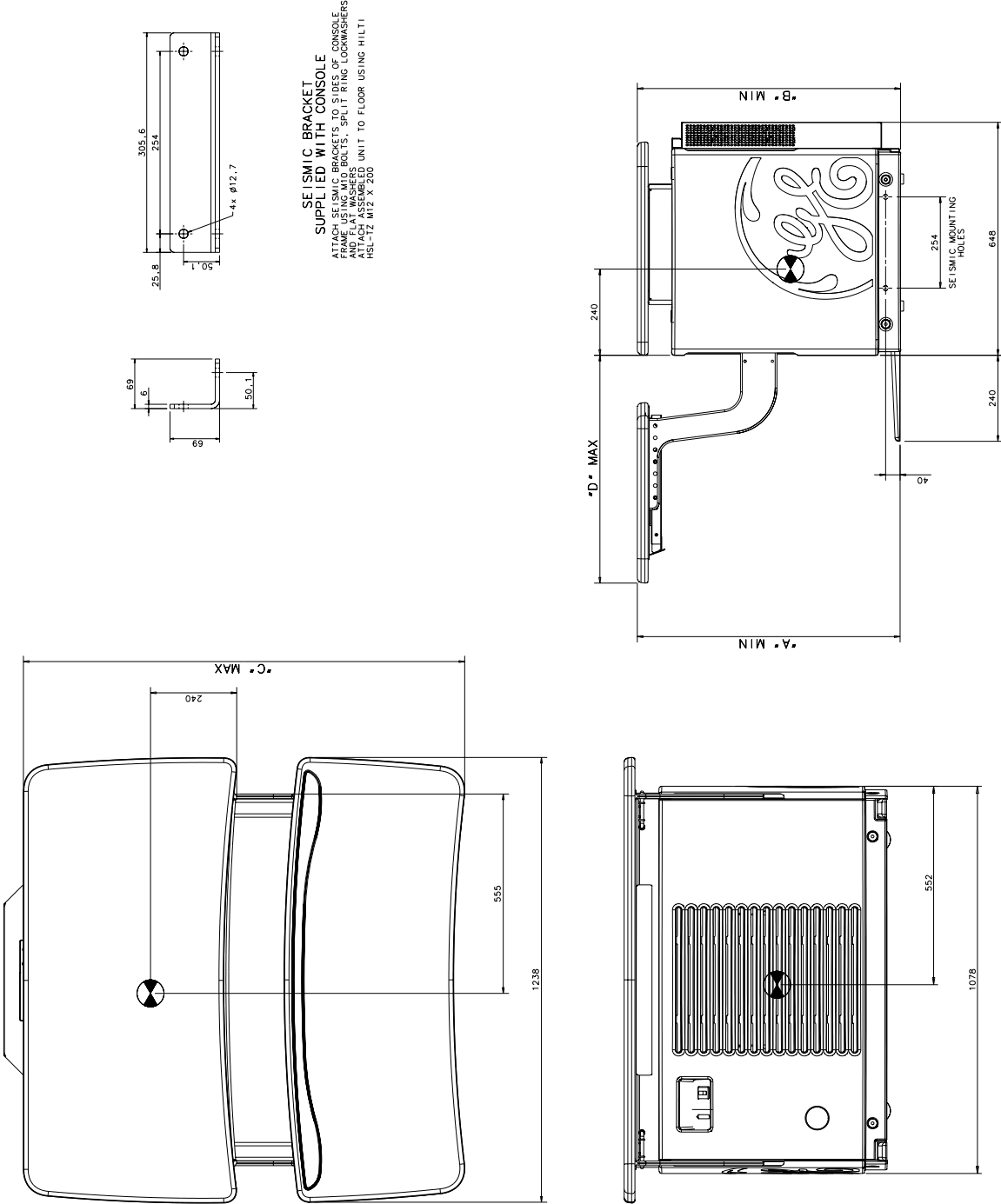


Figure 6-6 Operator's Console

Chapter 7

Delivery Data



WARNING SOME ASSEMBLIES ARE TOP-HEAVY. BE CAREFUL NOT TO TIP.

Section 1.0 Van Delivery

The CT system is packed for van shipment with minimum tear-down of components. It consists of approximately 20 shipping containers, which include dollies, skids and boxes without skids.

Section 2.0 Delivery/Shipping Requirements & Considerations

The LightSpeed 5.X System is not designed to tolerate excessive mishandling, including dropping, shock, vibration, tipping or hoisting.

The Gantry, Console, Table and PDU must NEVER be dropped. A drop from a height greater than one half inch ($\frac{1}{2}$ ") may induce structural damage to the frame or other major components. Damage resulting from a drop (e.g., bent frame, misalignment) may not be obvious until after system installation is complete.

To avoid dropping the Gantry, Dock to Dock shipment is recommended. Other methods are acceptable, provided that the system is not dropped or otherwise mishandled. For example, the system may be moved via flat-bed wrecker or by rolling it across SMOOTH sidewalks or other paved surfaces.

When moving Gantry off of a flat-bed wrecker, attach the straps to the lowest point possible on the dolly. Lower the Gantry at the slowest reasonable rate. See [Figure 7-1](#).

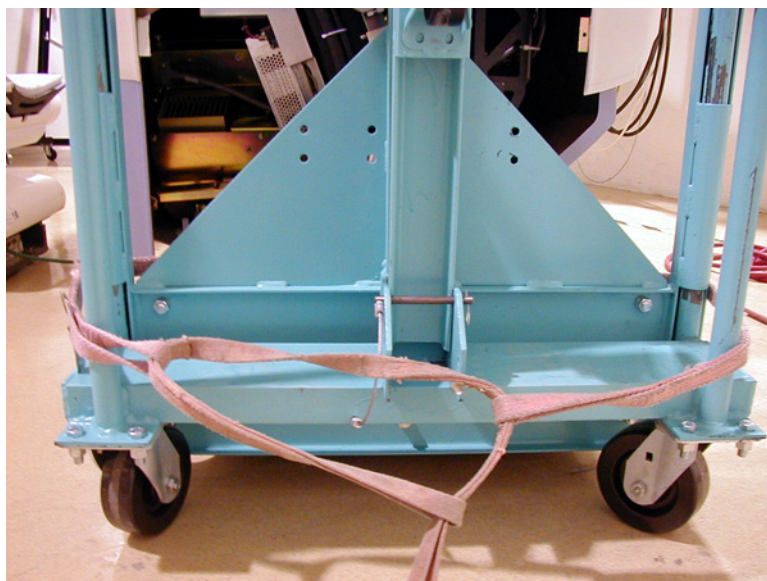


Figure 7-1 Gantry Strap Location

The LightSpeed 5.X System—including Gantry, Console, Table and PDU—is not designed to tolerate the excessive shock or vibration that may occur during unloading. For example, rolling the Console across a “washboard” style ramp may vibrate components to the extent of loosened or broken connections, etc. Damage resulting from shock or vibration (e.g., monitor, CD-ROM, hard-drive or octane failure) may not be evident until after system installation is complete.

All system components must remain upright at all times, and must not be tipped. Nor should the Gantry be hoisted. The Gantry should be moved by rolling it on its dollies only. Movement through hallways, doorways, elevators, etc., must be done without tipping or lifting the Gantry.

Protection for flooring along the move path (from dock to scan room) is advised.

Section 3.0

Site Environmental Considerations

3.1 Dust/Dirt Contamination

LightSpeed 5.X systems (consisting of: Console, PDU, Table and Gantry) are highly susceptible to airborne contaminants, especially concrete and drywall dust. Due to the possibility of contamination, these systems should NEVER be installed in a construction site. Any site with unfinished floors, walls or ceilings is considered a construction site, and is not suitable for system installation.



NOTICE The act of installing a GE CT scanner in a construction (i.e., unfinished) site will likely result in the following adverse effects:

- Increased installation time
- Decreased installation quality
- Increased scanner downtime, due to increased service calls

3.2 Chemical Contamination

Wet film processors must never be installed in the same room as the scanner, due to the possibility of chemical contamination of LightSpeed 5.X CT scanner components. Such chemicals can contribute to increased equipment failures, increased system downtime, and decreased reliability. Film processor equipment installation must meet the manufacturer’s requirements (e.g. ventilation specifications) and all applicable national and local codes. Also, consideration’s should be given to the location of this equipment and chemical fumes relative to human contact as it relates to locating this equipment and chemicals in the control room.

Section 4.0 Storage Requirements

SHORT TERM STORAGE (LESS THAN 6 MONTHS)

If the CT system is to be stored before installation, store in a temperature and humidity controlled warehouse. Protect from weather, dirt and dust.

Meeting these requirements prevents rust and corrosion from forming on bearing surfaces due to condensation.

- Storage temperature should not exceed 40° to +80° F (4° to +27° C).
- Maintain relative humidity (non-condensing) between 20% and 60%.
- Maximum relative humidity rate of change is 5%/hr.
- The maximum temperature rate of change is 5° F/hr. (3° C/hr.)
- Air pressure should be between 700hPa and 1060hPa.



NOTICE Between delivery is considered short-term storage.
Van storage must meet the same specifications as above.

LONG TERM STORAGE (6 MONTHS OR MORE)

If the CT system is to be stored before installation, store in a temperature and humidity controlled warehouse. Protect from weather, dirt and dust.

Meeting these requirements prevents rust and corrosion from forming on bearing surfaces due to condensation.

- Storage temperature should not exceed 50° to +90° F (10° to +32° C).
- Maintain relative humidity (non-condensing) between 20% and 70%.



Figure 7-2 Package Symbols (Storage)

Section 5.0 Extreme Temperature Transportation and Deliveries

Extreme temperatures should be avoided during system transportation and delivery.

Extreme temperatures are defined as below Zero degrees Fahrenheit (-18° C) or above 120 degrees Fahrenheit (49° C), without humidity control.

Section 6.0

System Transportation

When transporting the CT system, ensure that the system is not exposed, for an extended period of time, to temperatures or humidity outside of the following specifications.

Temperature: 0° to +120° F (-18° to +49° C)

Humidity: 0% to 80%



NOTICE

Component Freezing occurs if CT system is exposed to temperatures below 0° F (-18° C) for a period longer than two days.

Allow a minimum of 12 hours for the CT system to adjust to ambient room temperature, prior to installation.

Section 7.0

Gantry Considerations

The gantry is shipped with most covers installed. The assembly is mounted between two dollies. See [Figure 7-3](#). Two side rails are bolted to the dollies to stabilize dollies and protect gantry. Use dolly elevating casters to lift gantry off its base and roll it into position.



Figure 7-3 Gantry with Shipping Dollies and Side Rails

Door Openings. Clear door openings for moving equipment into building must be 42 X 82 in. (1067 X 2083 mm) minimum. 8 ft. (2439 mm) corridor width is helpful.

Elevator requirements. To move gantry from receiving location to scanning room, consider elevator capacity and size. By removing side rails and one dolly after gantry is placed in elevator, gantry width/length and elevator depth requirements are reduced. Contact a representative of elevator manufacturer if gantry weight exceeds elevator capacity.

CONFIGURATION	LENGTH IN (CM)	WIDTH IN (CM)	HEIGHT IN (CM)
Dollies On, Side Rails On	114 (290)	51 (129)	79 (200)
Dollies On, Side Rails Removed	114 (290)	39.4 (100)	79 (200)
Dollies Off, Covers Off	81 (206)	34 (86)	73 (185)

Table 7-1 Size of Gantry & Dollies, with and without Side Rails

Minimum hallway and door size for gantry with covers and dollies attached, but side rails removed, is 40 inches (101.6 cm).

For alternative lifting arrangements and instructions, contact GE Installation Support Services.

Dollies: Typically, dollies are used on the gantry, table, and console for domestic shipments. Once the gantry, table and console are installed at the site, return dollies to GE using the shipping document located in Box #1.

Dollies can be purchased for international shipments (B7850LD) to be used at the customer site. After the system has been removed from the crates, dollies shipped with international shipments **only** are not to be shipped back to GE in Milwaukee, WI, USA, but to be retained to the local GE office or warehouse.

Zero clearance dollies are available (P/N 2128404).

Section 8.0

Table Considerations

The table is shipped without side covers installed. Covers are shipped in four separate boxes. The table is mounted between two dollies.

Table shipping dimensions are 101" (2565 mm) long, 32" (81 mm) wide, and 47" (119 mm) high.

Section 9.0

Console Considerations

NEW INFORMATION -- PLEASE REVIEW THIS SECTION

The console is shipped without the keyboard table installed. The keyboard table is shipped with the console.

- The console is shipped on a skid equipped with ramps for unloading.
- Do not remove the console from the shipping skid until it is in the CT equipment room.
- Console shipping dimensions (on the shipping skid) are 40" (1016 mm) deep, 53" (1346 mm) wide, and 41" (1041 mm) high.

Note: The dimensions of the console alone (as shipped) are 34.95" (888mm) deep, 48.74" (1238mm) wide, and 29.5" (750mm) high.



NOTICE Do not lift the console by the monitor table top to remove from the shipping skid.



Figure 7-4 Console ready to be unloaded from shipping skid.

Chapter 8

Power Requirements

Section 1.0

Introduction

Note: Power requirements for LightSpeed 5.X differ from those for LightSpeed 4.X and previous systems.

The power distribution unit (PDU) supplied with the LightSpeed 5.0 system transforms and distributes power to all system components. The PDU is the only power entry point required to operate system.

To minimize voltage regulation effects, power wiring between the facility main distribution panel and the PDU should be kept as short as possible.

When routing the power wiring all three phase wires and ground must be run in the same conduit or raceway duct. Power wires should be routed separately from system control and signal cables, using a separate conduit or trough in raceway duct.

Metallic conduit, floor duct or surface raceway may be used for running cables, depending upon local codes and practices. However, cable passageways should be large enough to install any cable with all other cables already installed. Use of non-metallic conduit is not recommended.

Section 2.0

System Input Power

2.1 Power Source Configuration

The LightSpeed 5.0 CT Scanner is designed to operate on a three-phase, four-wire delta or wye power source. Although a solidly grounded wye source is preferred, an impedance grounded wye source may be used. The neutral wire does not need to be run to the system, i.e., four-wire connection.

If the only power source available is a delta configuration the customer must provide a wye connected grounding transformer or autotransformer at the source. Corner or mid-phase grounded delta sources are NOT recommended.

A dedicated feeder from the nearest Main Distribution Panel (MDP) should be used to supply power to the scanner. In accordance with the National Electric Code (U.S.) and similar applicable national and local codes, a protective disconnect device must be provided in the power line supplying the PDU. It must be located within 32 ft (10 m) of the PDU, visible to PDU service personnel, and must have "lockout / tagout" provisions. This disconnect device is identified as "A1" in the interconnection schematic diagrams.

2.2 Rating

The LightSpeed 5.0 system operates on three-phase power meeting the following specifications.

Voltage	200 to 480 VAC
Capacity	90 kVA (75kVA)
Frequency	50 or 60 Hz +/- 3 Hz

- Maximum power demand = 90 kVA (75kVA) @ 0.85 PF at a selected technique of 140 kV, 380 mA (300mA).
- Average (continuous) power demand at maximum duty cycle = 20 kVA.

The "A1" disconnect device referenced above must provide overcurrent protection for the system and facilitate multi-point remote "Emergency Off" control of system power. A disconnect utilizing undervoltage release control is preferred over shunt trip devices. The rating of the "A1" disconnect device depends on the nominal line voltage at the site. Refer to [Section 3.0: "Recommended Power Distribution System"](#), for minimum rating requirements and suggested disconnect devices.



NOTICE The electrical rating is described on the system rating label attached on the gantry; not on the PDU.

2.3 Regulation

Total load regulation as measured at the PDU input terminals must not exceed 6%. The capacity of the facility transformer and size & length of feeder wires directly affect the load regulation presented to the system. Refer to [Section 3.0: "Recommended Power Distribution System"](#), for recommended single-unit installation specifics.

2.4 Phase Imbalance

The difference between the highest line-to-line voltage and lowest line-to-line voltage must not exceed 2% of the lowest line-to-line voltage.

2.5 Sags, Surges & Transients

Sags and surges of the power line must not exceed the absolute range limits shown in Table 8-1. Maximum transient voltages should be limited to 1500V peak.

2.6 Grounding

Metal conduit, raceway or the armor of armored cable used to power the system should be bonded to the PDU cabinet. However, in addition to such mechanical grounding, a dedicated 1/0 (55 mm²) or larger insulated copper ground wire must be run with the phase wires from the main distribution panel to the PDU.

Note: The shield or armor of armored cable is not sufficient for this purpose.

The ground wire should be bonded to intermediate distribution panels through which it passes in accordance with local codes. The resistance between the PDU ground and the facility earth ground must not exceed 0.5 ohm. In addition, the total resistance between the PDU ground and earth must not exceed 2 ohms.

Section 3.0

Recommended Power Distribution System

A dedicated feeder run from the facility main isolation transformer is recommended to power the LightSpeed 5.0 CT scanner. If the scanner must be powered from an existing distribution transformer and secondary feeder, such as the equipment distribution panel of an X-ray department, installation with other X-Ray equipment that use rapid film changers should be avoided. These changers use a large number of high powered, closely spaced exposures, which may coincide with the CT scan and produce image artifacts.

If a dedicated distribution transformer is provided for the scanner, the minimum recommended transformer size is 112.5 kVA, rated 2.4% regulation at unity power factor. For this configuration, the minimum recommended feeder size and overcurrent protection device based on line voltage is shown in [Table 8-3](#) Minimum Feeder Wire Size.

In all cases, qualified personnel must verify that the transformer and feeder, at point of take-off, plus the run to the LightSpeed 5.0 CT scanner meet all the requirements stated in this document.

SYSTEM CHARACTERISTICS:

[\(xxx\):The value for 75kVA.](#)

- Maximum power demand = 90kVA (75kVA) @ 0.85 PF: at a Selected Technique of 140 kV, 380 mA (300mA).
- Continuous (average) power demand at maximum duty cycle = 20kVA (16.7kVA).
- Maximum allowable total source regulation is 6%.
- Minimum recommended transformer size: 112.5 kVA (93.75kVA), with 2.4% rated regulation at unity power factor. Resultant maximum allowable feeder regulation is 3.4%.

The nominal line voltage must fall within one of the ranges listed below

Nominal Line Voltage	200	220	240	380	400	420	440	460	480
Hi-Line Limit, +10%	220	242	264	418	440	462	484	506	528
Lo-Line Limit, -10%	180	198	216	342	360	378	396	414	432
Continuous Line Current	58	52	48	30	29	27	26	25	24
Momentary Line Current	260	236	217	137	130	124	118	113	108
Maximum Line Current	289	262	241	152	144	137	131	126	120
Minimum Recommended Circuit Protection Rating	150	150	150	110	110	100	100	90	90

Table 8-1 Nominal Line Voltage

OPTION	CAT NUM
225 kVA Transformer	E4500AW
Isolation Transformer	E4500BC

Table 8-2 Purchasable Options

FEEDER LENGTH (MDA TO A1) FEET (METERS)	MINIMUM FEEDER WIRE SIZE, AWG OR MCM (SQ. MM)/ VAC								
	200 VAC	220 VAC	240 VAC	380 VAC	400 VAC	420 VAC	440 VAC	460 VAC	480 VAC
50 (15)	1/0 (55)	1/0 (55)	1/0 (55)	2 (35)	2 (35)	3 (30)	3 (30)	3 (30)	3 (30)
100 (30)	2/0 (70)	1/0 (55)	1/0 (55)	2 (35)	2 (35)	3 (30)	3 (30)	3 (30)	3 (30)
150 (46)	4/0 (100)	3/0 (85)	2/0 (70)	2 (35)	2 (35)	3 (30)	3 (30)	3 (30)	3 (30)
200 (61)	5/0 (125)	4/0 (100)	4/0 (100)	2 (35)	2 (35)	3 (30)	3 (30)	3 (30)	3 (30)
250 (76)	6/0 (170)	5/0 (125)	5/0 (125)	1 (45)	1 (45)	2 (35)	2 (35)	2 (35)	3 (30)
300 (91)	7/0 (215)	6/0 (170)	5/0 (125)	1/0 (55)	1/0 (55)	1 (45)	1 (45)	2 (35)	2 (35)
350 (107)	8/0 (275)	7/0 (215)	6/0 (170)	2/0 (70)	1/0 (55)	1/0 (55)	1 (45)	1 (45)	1 (45)
400 (122)	8/0 (275)	7/0 (215)	7/0 (215)	2/0 (70)	2/0 (70)	1/0 (55)	1/0 (55)	1/0 (55)	1 (45)

Table 8-3 Minimum Feeder Wire Size

SUB-FEEDER LENGTH (A1 TO PM) FEET (METERS)	MINIMUM SUB-FEEDER WIRE, AWG OR MCM (SQ. MM)								
	200 VAC	220 VAC	240 VAC	380 VAC	400 VAC	420 VAC	440 VAC	460 VAC	480 VAC
32(9.7536)	1/0 (55)	1/0 (55)	1/0 (55)	2 (35)	2 (35)	3 (30)	3 (30)	3 (30)	3 (30)

Table 8-4 Minimum Feeder Wire Size

- 1.) [Table 8-1](#), [Table 8-3](#), and [Table 8-4](#) above are based on the use of copper wire, rated 75 C and run in steel conduit. Ampacity is determined in accordance with the National Electrical Code (NFPA 70), Table 310-16 (2002)
- 2.) The minimum feeder size is determined by the ampacity of the circuit protection device listed above, except where a larger size is necessary to meet total source regulation limits.
- 3.) A 1/0 (55 sq. mm) ground wire is recommended in all cases.

Section 4.0

Power Source Monitoring

A power audit should be conducted at the customer site prior to installation of the LightSpeed 5.0 CT Scanner. During the audit, a power line analyzer should be used to check the site power for average line voltage, frequency, sags, surges, and transient impulse activity. A minimum ten-day period that includes two weekends should be monitored to encompass several days of normal use. Verify "brown-out" (low voltage) conditions, which may occur during summer months, will not exceed the selected allowable range shown in [Table 7-1](#).

Some analyzer models, which are suitable for power line monitoring, are:

- Dranetz Model 658
- Dranetz Model 656
- BMI 3630

The data and any known site history of previous power problems with other X-ray systems or computer installations should be analyzed and reviewed with your GE Power and Ground representative.

Section 5.0 Ground System

The LightSpeed 5.0 CT Scanner has been designed to use an equal potential grounding system. The required ground system is shown in [Figure 7-1](#). There are three primary grounding points:

- A system power ground point located in the PDU.
- A reference ground point located between gantry and table base.
- A patient ground point located at the front of the table base.

All exposed metal surfaces in the patient vicinity are grounded to the reference ground point. For additional information, refer to Electrical Safety Equipment, Direction 46-014505.

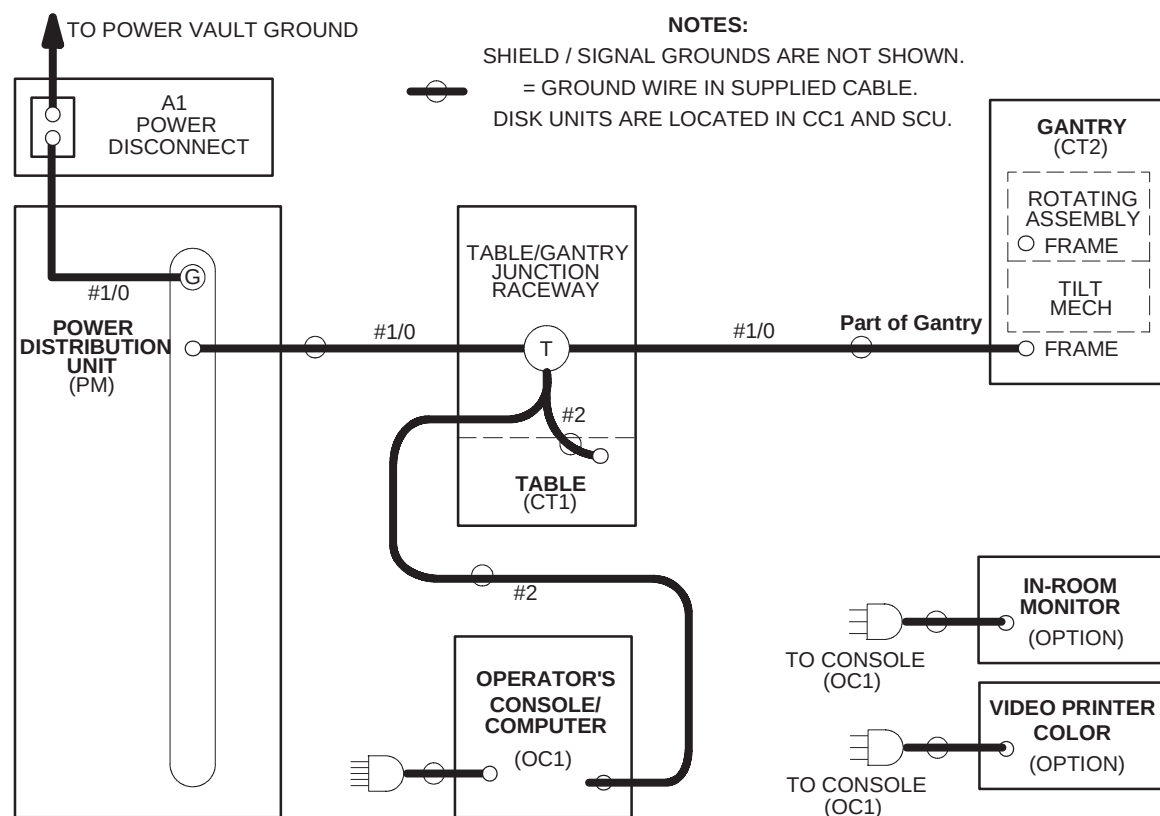


Figure 8-1 System Ground Map

Chapter 9

Interconnection Data

Section 1.0 Introduction

Figure 9-3 shows interconnection runs for a 50/60 Hz system.

Table 9-1 shows component designators for supplied equipment and options and wall power outlets.

Table 9-5 lists customer-installed wiring and supplied cables. Actual length of each run is less than the length of supplied cables to allow for routing inside equipment. Cable diameters and sizes of connectors are provided to aid in sizing conduit and access plates.

Table 9-2 and Table 9-3 list details for connection to LightSpeed Series equipment, using standard (short) length and non-standard (long) length cables, respectively. Details are listed for the following types of runs as appropriate:

- Flush-floor duct
- Computer floor
- Through-wall bushing
- Junction box
- Surface floor duct
- Through-floor duct
- Wall duct
- Conduit

Need for additional junction boxes is minimized by use of either a cable raceway system or a raised computer floor. LightSpeed Series systems use prefabricated cables with large plugs. Therefore, conduit or pipe is not recommended for cable runs.

The catalog number of the non-standard (long) length cable set is B7816PH (H-Power Long Cable Set). Order this, if required for your site.

Note: Use dry cleansing for electro components.

Section 2.0 Component Designators

DESIGNATOR	APPLIES TO	SOURCE
A1	Primary power disconnect	Contractor supplied
CT1	Patient table	System
CT2	Gantry	System
ITL	InSite telephone lines	Contractor supplied
LP	Line printer	Option
OC1	Operator's console/computer	System
PM	Power distribution unit	System
SEO	System emergency off	Contractor supplied
SM	Slave monitor	Option
WL	"X-ray on" warning light	Contractor supplied
DS	Door Interlock Switch	Contractor supplied
XCVR	Ethernet transceiver	System

Table 9-1 Component Designators

Section 3.0 Interconnect Runs, Wiring and Cables

3.1 GEMS Supplied (Standard Length)

RUN #	LENGTH, ACTUAL (USABLE)		PART #	DESCRIPTION	UL CABLE INFORMATION								PULL SIZE MM (INCHES)
	ft	m			UL Style	Flam. Rating	Voltage Rating	Voltage Actual	Temp. Rating (C)	Dia. mm (inch)	# of Cond	Size AWG	
050	28 (20)	8.5 (6.1)	2343529-2	HVDC, PDU to Gantry	2587	FT4	600	+ & - 350VDC	90	19 (.751)	3	(2) 4 (1) 8	22 (.87) Dia
051	28 (20)	8.5 (6.1)	2343530-2	HVAC, PDU to Gantry	2587	FT4	600	440Y/254	90	15.3 (.604)	4	14	11.2 (.44) Dia
052	28 (20)	8.5 (6.1)	2343528-2	LVAC, PDU to Gantry	2587	FT4	600	208Y/120	90	13.8 (.542)	5	8	56.4 (2.22) Dia
053	65 (60)	19.8 (18.3)	2343531-2	LVAC, PDU to Console	2587	FT4	600	120VAC	90	12.3 (.483)	3	10	56.4 (2.22) Dia
054			n/a	LVAC, Gantry to Table	1015		600	120VAC			3	14	
055	43 (35)	13.2 (10.67)	2371450-2	Ground, PDU to Raceway	1284	VW-1 (FT-1)	600	0	105	15.5 (.608)	1	1/0	15.8 (.62) Dia
056	71 (60)	21.7 (18.3)	2371450-4	Ground, Raceway to Console	1283	VW-1 (FT-1)	600	0	105	11.9 (.467)	1	2	12.2 (.48) Dia
100	32.5 (20)	9.9 (6.1)	2333152-2	Signal, Gantry MSUB to PDU		FT-4	300	<30VDC	80	11.2 (.440)	25	22	17 x 58 (.68 x 2.30) 19 x 51 (.75 x 2.01)
101	71 (60)	21.7 (18.3)	2333150-2	Signal, Gantry MSUB to OC		FT-4	300	<30VDC	80	11.2 (.440)	25	22	17 x 58 (.68 x 2.30) 19 x 51 (.75 x 2.01)
102	68 (60)	20.7 (18.3)	2352714-3	Signal (LAN), Gantry to OC			1900	<30VDC		5.9 (.234)	8	24	15 (.59) Dia
103	68 (60)	20.7 (18.3)	2117848-7	Fiber Optic, Gantry to OC			N/A	N/A			1	N/A	10 (.39) Dia
104			n/a	Signal, Gantry to Table		FT-4	300		80		25	22	

Table 9-2 GEMS Supplied Cables (Standard Run) - UL Information

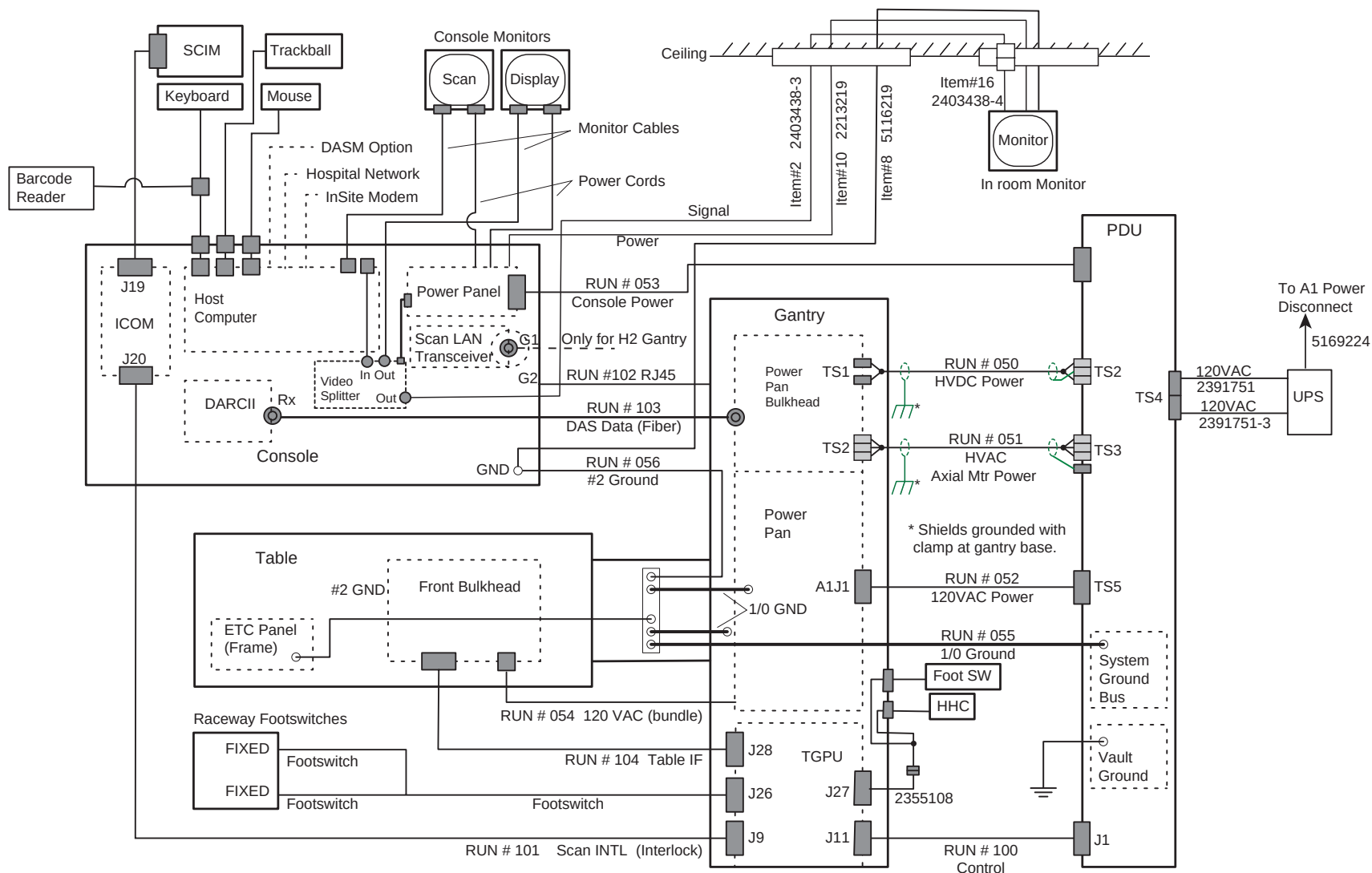


Figure 9-1 System Interconnect Diagram

3.2 GEMS Supplied (Optional, Long Run)

RUN #	LENGTH, ACTUAL (USABLE)		PART #	DESCRIPTION	UL CABLE INFORMATION								PULL SIZE MM (INCHES)
	ft	m			UL Style	Flam. Rating	Voltage Rating	Voltage Actual	Temp. Rating (C)	Dia. mm (inch)	# of Cond	Size AWG	
050	63 (55)	19.3 (16.76)	2343529	HVDC, PDU to Gantry	2587	FT4	600	+ & - 350VDC	90	19 (.751)	3	(2) 4 (1) 8	22 (.87) Dia
051	62.5 (55)	19 (16.76)	2343530	HVAC, PDU to Gantry	2587	FT4	600	440Y/254	90	15.3 (.604)	4	14	11.2 (.44) Dia
052	60 (55)	18.5 (16.76)	2343528	LVAC, PDU to Gantry	2587	FT4	600	208Y/120	90	13.8 (.542)	5	8	56.4 (2.22) Dia
053	80 (75)	24.5 (22.86)	2343531	LVAC, PDU to Console	2587	FT4	600	120VAC	90	12.3 (.483)	3	10	56.4 (2.22) Dia
055	63 (55)	19.3 (16.76)	2371450	Ground, PDU to Raceway	1284	VW-1 (FT-1)	600	0	105	15.5 (.608)	1	1/0	15.8 (.62) Dia
056	83 (75)	25.5 (22.86)	2371450-3	Ground, Raceway to Console	1283	VW-1 (FT-1)	600	0	105	11.9 (.467)	1	2	12.2 (.48) Dia
100	63 (55)	19.3 (16.76)	2333152	Signal, Gantry MSUB to PDU		FT-4	300	<30VDC	80	11.2 (.440)	25	22	17 x 58 (.68 x 2.30) 19 x 51 (.75 x 2.01)
101	83 (75)	25.5 (22.86)	2333150	Signal, Gantry MSUB to OC		FT-4	300	<30VDC	80	11.2 (.440)	25	22	17 x 58 (.68 x 2.30) 19 x 51 (.75 x 2.01)
102	80 (75)	24.3 (22.86)	2373436-1	Signal (LAN), Gantry to OC			1900	<30VDC		5.9 (.234)	8	24	15 (.59) Dia
103	80 (75)	24.3 (22.86)	2117848-2	Fiber Optic, Gantry to OC			N/A	N/A			1	N/A	10 (.39) Dia

Table 9-3 GEMS Supplied Cables (Optional, Long Run) - UL Information

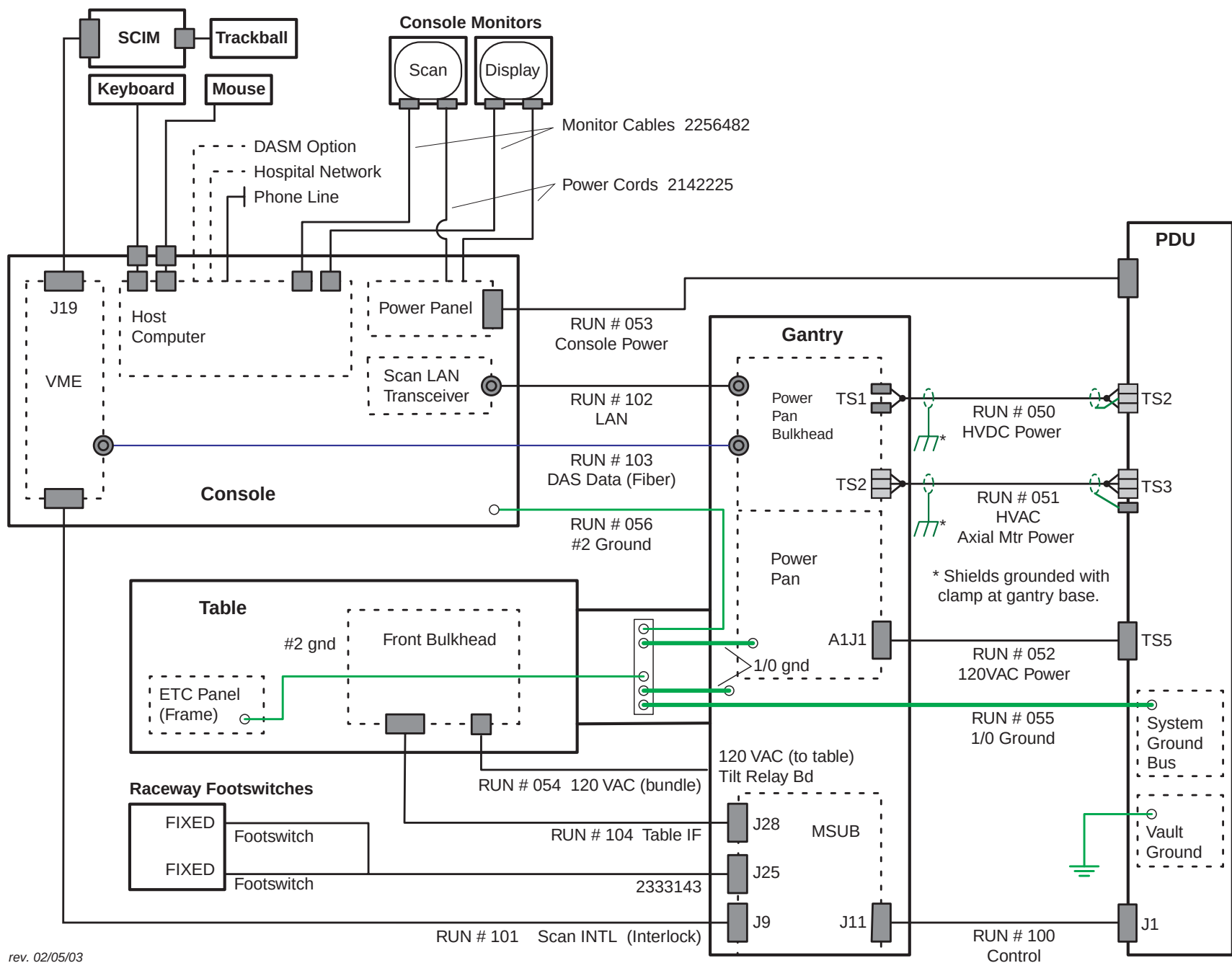


Figure 9-2 System Interconnect Diagram

rev. 02/05/03

3.3 GEMS Supplied (Cables of Options)

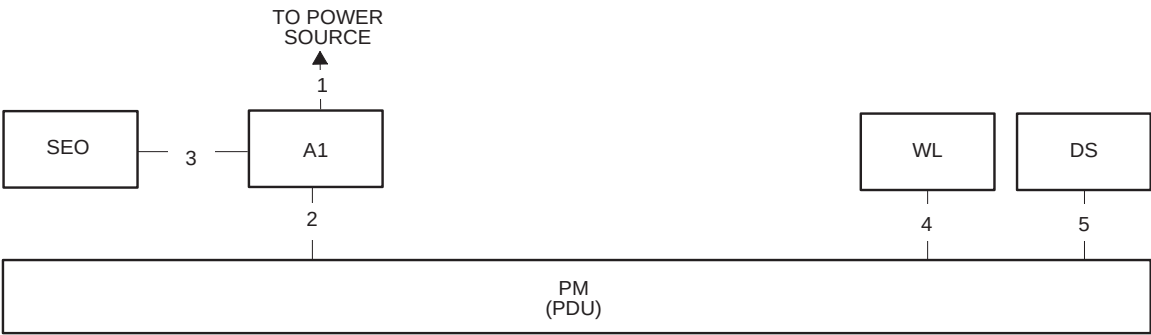
OPTION	LENGTH, ACTUAL (USABLE)		PART #	DESCRIPTION	UL CABLE INFORMATION								PULL SIZE MM (INCHES)
	ft	m			UL Style	Flam. Rating	Voltage Rating	Voltage Actual	Temp. Rating (C)	Dia. mm (inch)	# of Cond	Size AWG	
Fluoro	75	22.9	2403438-3	5 BNC MALE TO HD 15 MALE 75 FEET		FT4		1Vp-p	75	9.1 (0.358)	5	26	
	70	21.2	2213219	POWER CABLE FOR LCD-CONSOLE TO LCD		FT1	120	120VAC	105	9.3 (0.366)	3	14	
	71	21.5	5116219	Grounding Cable For LCD Console To LCD	1015	VW-1	600	0V	105		1	8	
	15	4.6	2403438-4	HD 15 FEMALE TO HD 15 MALE 15 FEET				1Vp-p	60	8.0 (0.315)	5	26	
	1.3	0.4	2355108	JUMPER CABLE FOR ADAPTING 2286150 TO WORK WITH H-POWER MSUB							8	22	
UPS	15	4.6	2391751	POWER CABLE, NGPDU TO UPS	2587	FT4	600	208VAC	90	5.8 (0.228)	5	8	
	15	4.6	2391751-3	POWER CABLE, UPS DISCONNECT PANEL TO NGPDU	2587	FT4	600	208VAC	90	5.8 (0.228)	4	8	
	45	13.6	5169224	UPS CONTROL CABLE	2587	FT4	600	120VAC	90	10.3 (0.406)	5	18	

Table 9-4 GEMS Supplied Cables for Options - UL Information

3.4 Contractor (Customer) Supplied

CUSTOMER INSTALLED WIRING		DESCRIPTION	CABLES SUPPLIED			PLUG PULLING DIMENSIONS		WIRE & CABLE PIGTAILS FT. (M.)	
QTY	SIZE AWG (MM ²)		PART NO	LENGTH FT. (M.)	DIA. IN (MM)	FROM	TO	FROM	TO
RUN NO. 1 FROM PRIMARY POWER SOURCE TO FACILITY DISCONNECT (POWER SOURCE - A1)									
Maximum Run Length *									
3	*	POWER						3 (1)	3(1)
1	1/0 (50)	GROUND						3 (1)	3 (1)
RUN NO. 2 FROM FACILITY DISCONNECT TO POWER MODULE (A1 - PM) MAXIMUM RUN LENGTH *									
3	*	POWER						3 (1)	3(1)
1	1/0 (50)	GROUND						3 (1)	3 (1)
1	*	NEUTRAL						3 (1)	3 (1)
RUN NO. 3 FROM FACILITY DISCONNECT TO SYSTEM EMERGENCY OFF (A1 - SEO)									
2	14 (2)	POWER						6 (2)	6 (2)
1	14 (2)	GROUND						6 (2)	6 (2)
RUN NO. 4 POWER MODULE TO WARNING LIGHT CONTROL (PM - WL)									
2	14 (2)	WARNING LIGHT 24 VOLT CONTROL A3J2-1,2,3,4							
RUN NO. 5 POWER MODULE TO SCAN ROOM DOOR INTERLOCK (PM - DOOR SWITCH)									
2	14 (2)	SCAN ROOM DOOR INTER LOCK A3J6-1,2							
*	REFER TO Table 8-4 on page 93 FOR AWG (MM2) WIRE SIZES.								

Table 9-5 Runs 1, 2, 3, 4 and 5 Connections



NOTES:

- 1) Used for remote diagnostics - Option
- 2) Refer to the appropriate Pre-installation / Installation documents for the Laser Camera
- 3) Category 5 cable. Use one of the following patch cords:

CAT Num	GE Part Num	
Length		
K9000WB	2215028-10	20 m
K9000KP	2215028-5	10 m
K9000JR	2215028-4	5 m

- 4) In order to avoid any violation of each National Regulation (NEC in USA, CCC in China, etc.), use of the complied cable/wire is recommended. For China market, China end-user shall purchase the power supply cable that has the CCC mark.

Only one phone connection is required for the system.

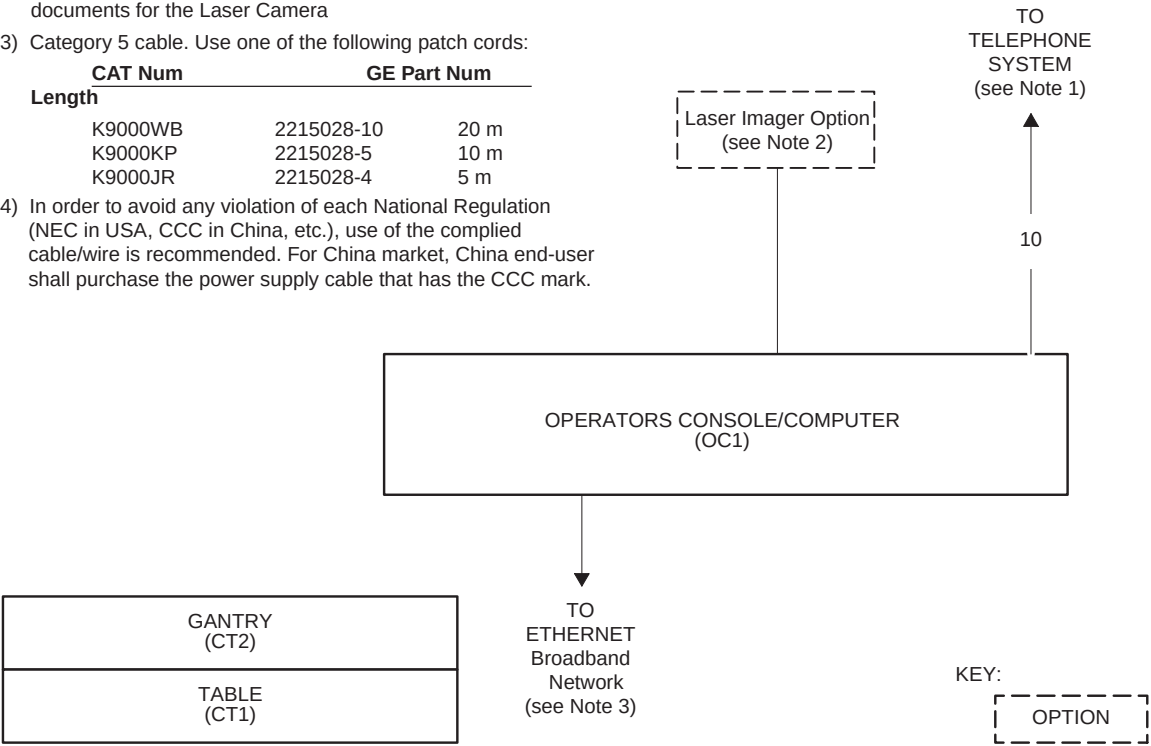


Figure 9-3 Interconnection Runs

Section 4.0

Contractor Supplied Components

REFERENCE	ASSOCIATED EQUIPMENT	MATERIAL/LABOR SUPPLIED BY CUSTOMER CONTRACTOR	USA VENDOR / CAT NO. GE CATALOG
A1	Fusible Disconnect and Magnetic Contactor	3 Pole, Magnetic Combination Contactor. With control transformer for 220V 50 Hz contactor coil, momentary contact start-stop push-button station, indicator light, multiple neutral block, multiple ground block and dual element fuses.	Source Locally or E4502JP.
ITL	In-suite Telephone Lines	Supply 2 voice-grade telephone lines. One line must be a direct number from outside the facility – do not route this line through a telephone switchboard. Telephone line operating charges are paid by customer.	
MFC	GE LaserCam HQ	See Chapter 10, Section 2.0 , for listings of Camera part numbers and descriptions.	
	System Components	Reference the system installation drawings supplied by Installation Support Services within your geographic area.	

Table 9-6 Contractor-Supplied Components

Section 5.0

Fuse

Item	number	qty	fru code	description/name
1	2351493	3.0	Yes	100A FUSE
2	2364059	2.0	Yes	GLASS FUSE
3	46-170021P50	2.0	Yes	FUSE 12 AMPS 250 VOLTS BUSSMAN MDA12 DUAL ELEMENT.
4	46-170021P15	2.0	Yes	Time Lag FUSE 8 AMPS 250 VOLTS (REVIEWED TK, 7/91).
5	2336517-2	2.0	Yes	FUSE 25 AMPS 700 VOLTS 2.5M SECONDS
6	46-170021P52	3.0	Yes	FUSE 3 AMPS 250 VOLTS (REVIEWED TK, 7/91).
7	46-170021P10	2.0	Yes	006.000A 0250V 3AG FAST UL/C (REVIEWED TK, 7/91).
8	46-170021P101	1.0	Yes	FUSE 20 AMPS 700 VOLTS 2~ LONG X 9/16~ DIA FAST BLOW
9	46-170021P106	2.0	Yes	FUSE 8 AMPS 250 VOLTS BUSSMANN ONLY
10	46-170021P31	2.0	Yes	1/2A, 250V SLO-BLO FUSE. TYPE 3AG, 1.25~ X 0.25~ GLASS BODY. (REVIEWED TK, 7/91).
11	2106993-5	2.0	Yes	FUSE 20 AMPS 500 VOLTS .3 SECONDS
12	46-327160P1	2.0	Yes	12.0A, 125VAC, DUAL ELEMENT TIME DLY FNM-12, 0.406 X 1.5 LG, FIBER BODY
13	2379651	1.0	Yes	FUSE - 700 V, 200 A, FAST SEMI
15	2238207-3	1.0	Yes	FUS FF 14X51 2A 700VDC C 50KA ULR
16	99183979	1.0	Yes	SLOW BLOW FUSE HCP 6X32 10A 250V UL

Table 9-7 P2030BE Fuse Kit HP60 90KVA (2385412 BOM, rev 2)

Section 6.0

UPS Interconnect



Figure 9-4 Typical UPS

Section 7.0

Typical Customer Supplied Wiring

7.1 Primary Power Disconnect

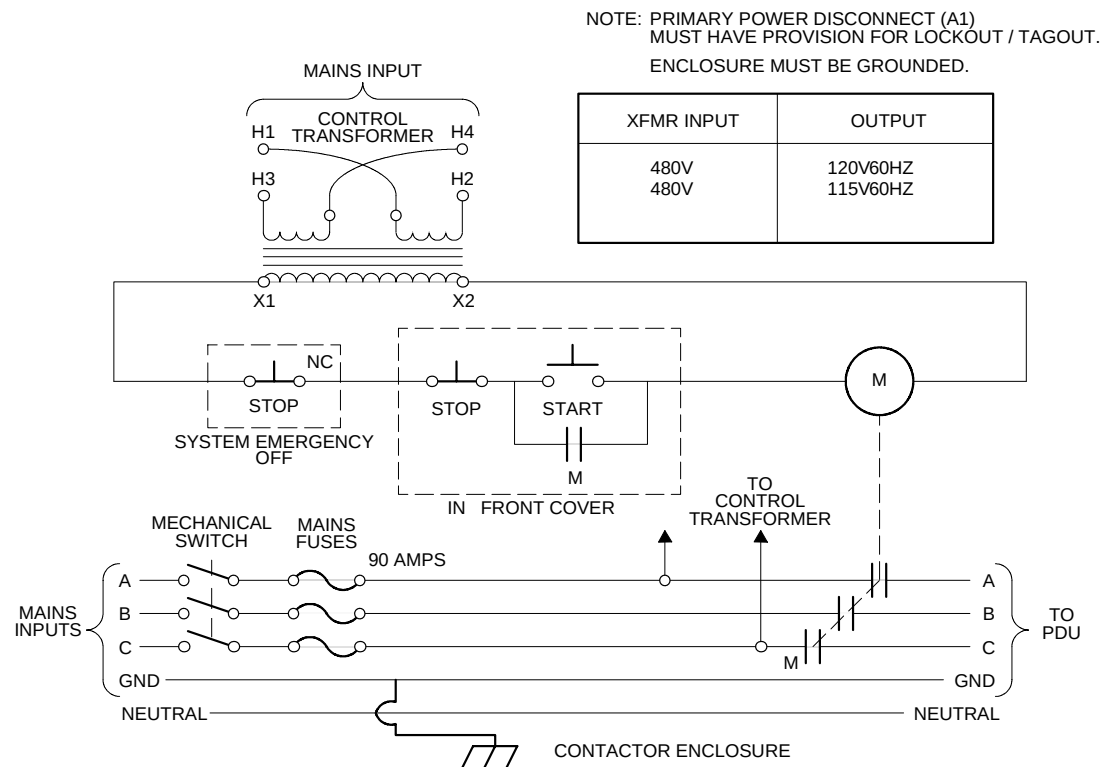


Figure 9-5 Primary Power Disconnect (A1) – Fusible Disconnect and Magnetic Contactor



Figure 9-6 Primary Power Disconnect (A1)

7.2 Scan Room Warning Light & Door Interlock

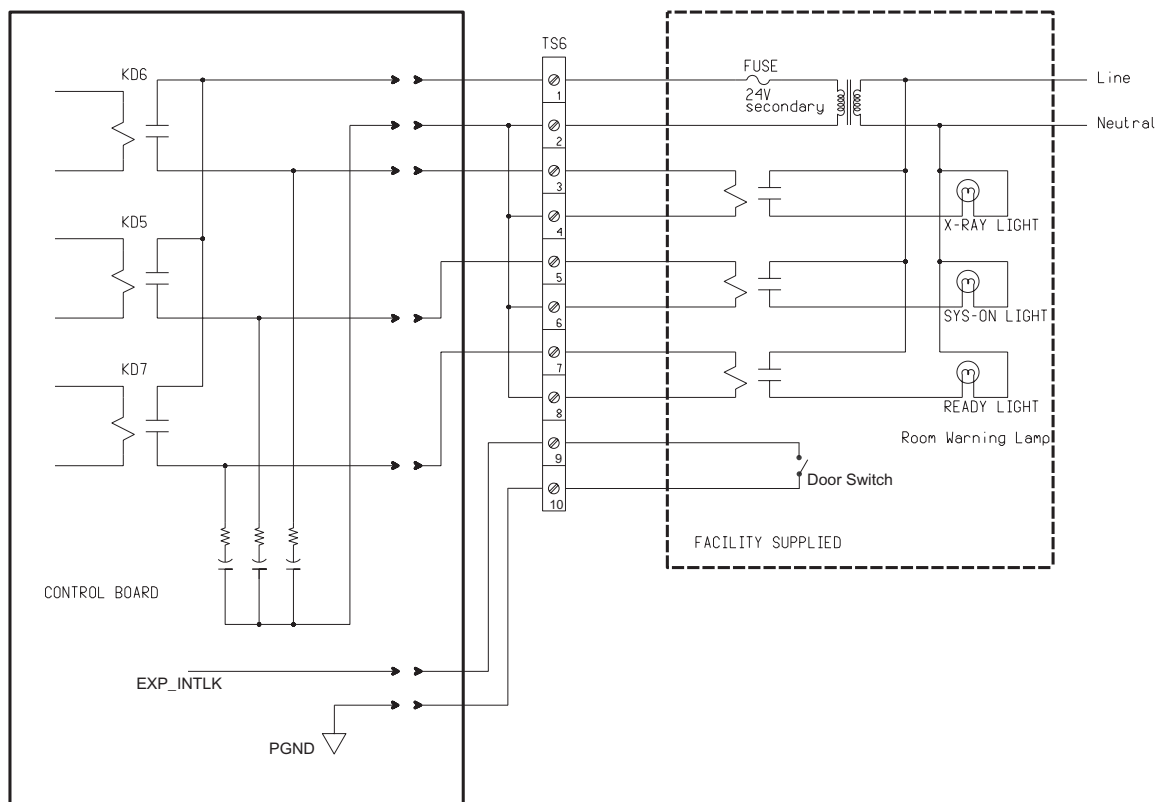


Figure 9-7 Typical TS6 Warning Light & Door Interlock Connections

Appendix A

Symbols

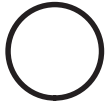
SYMBOL	PUBLICATION	DESCRIPTION
	417-5032	Alternating Current
	335-1	Three-phase Alternating Current
	335-1	Three-phase Alternating Current with neutral conductor
		Direct Current
	417-5019	Protective Earth (Ground)
	348	Attention, consult ACCOMPANYING DOCUMENTS
	417-5008	OFF (Power: disconnection from the mains)
	417-5007	ON (Power: connection to the mains)
		Warning, HIGH VOLTAGE
		Emergency Stop
		Type B

Table 9-8 Symbols

SYMBOL	PUBLICATION	DESCRIPTION
	417-5339	X-ray Source Assembly Emitting
	417-5009	Standby
		Start
		Table Set
		Abort
		Intercom
		(on Operator Console) Power On: light on Standby: light off

Table 9-8 Symbols



GE MEDICAL SYSTEMS

**GE MEDICAL SYSTEMS-AMERICAS: FAX 262.312.7434
3000 N. GRANDVIEW BLVD., WAUKESHA, WI 53188 U.S.A.**

**GE MEDICAL SYSTEMS-EUROPE: FAX 33.1.40.93.33.33
PARIS, FRANCE**

**GE MEDICAL SYSTEMS-ASIA: FAX 65.291.7006
SINGAPORE**

<http://www.gemedicalsystems.com>